





Computer-Integrated Surgery: Applications in Neurosurgery

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September 24, 2024

Disclosures

There are no financial or other conflicts of interest in relation to this presentation.



Outline

- Neurosurgery: An Overview
- Computer Integration to Modernize Neurosurgery
- Robotics in Neurosurgery



Outline

- **Neurosurgery: An Overview**
- Computer Integration to Modernize Neurosurgery
- Robotics in Neurosurgery



1889 – Johns Hopkins Hospital founded



Harvey Cushing



1896 – Surgical assistant to **William Halsted**

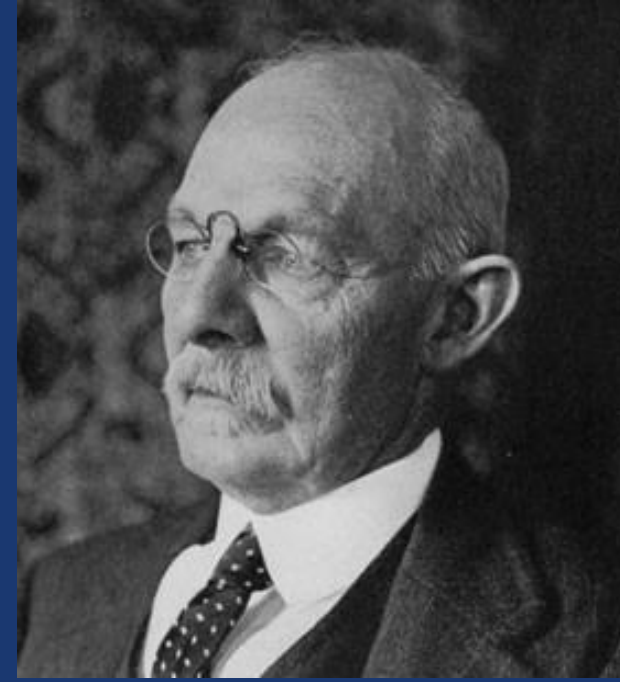
Osler



Cushing



Halsted

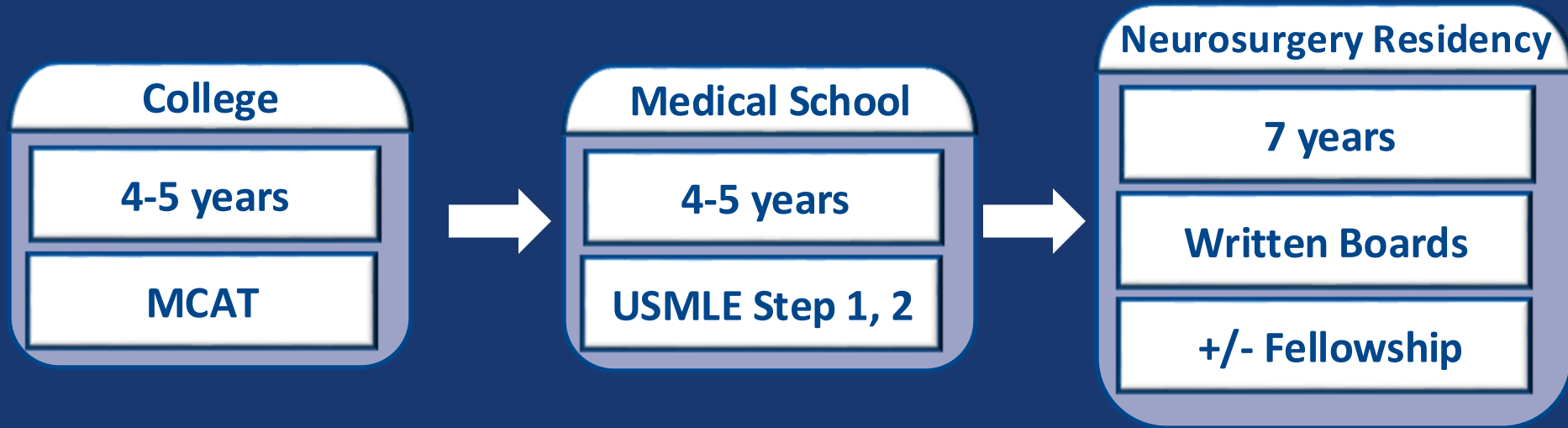


Harvey Cushing – founded modern
neurosurgery at Hopkins

The Department of Neurosurgery



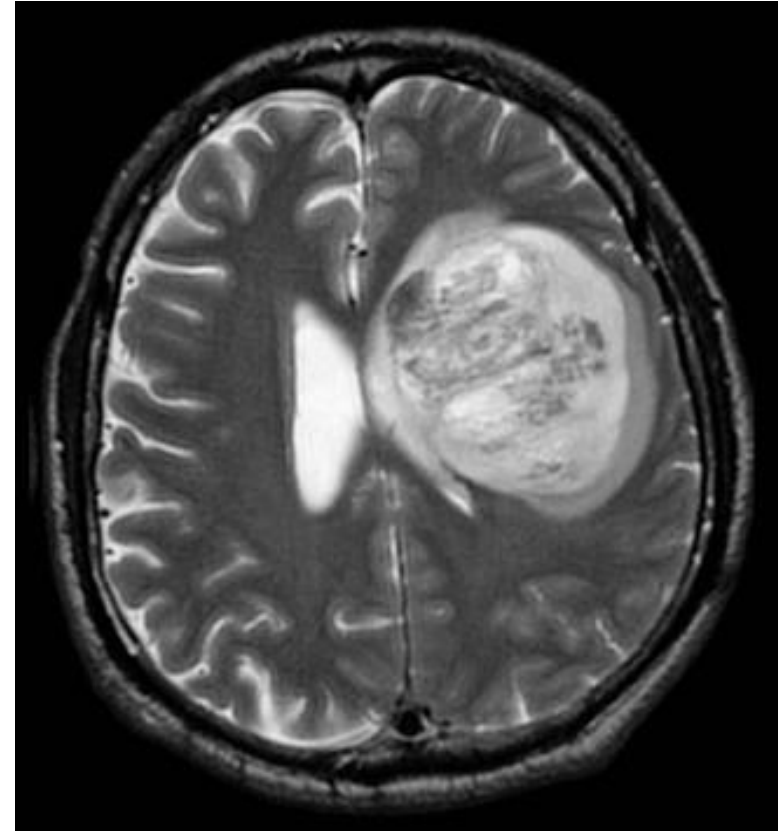
Path to becoming a neurosurgeon



Tumor – Neuro-Oncology

Comprehensive management of brain tumors.

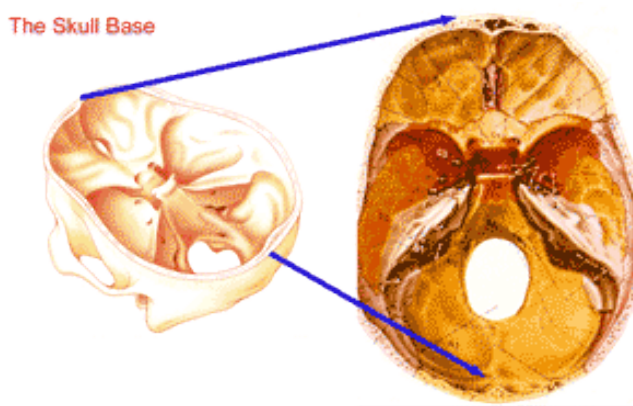
- Awake surgery
- Electrophysiological mapping
- Laser-induced thermal therapy
- Gamma Knife radiosurgery



Tumor – Skull Base

Emphasis on tumors arising along base or floor of skull

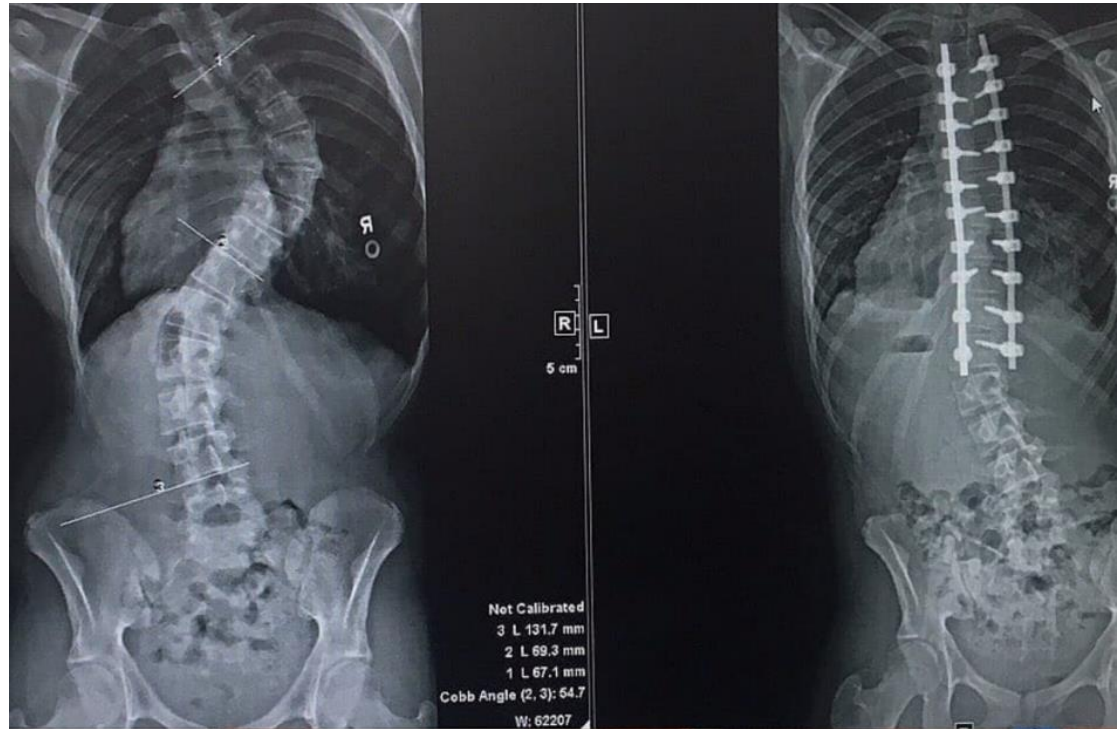
- Transcranial microsurgical approaches
- Endoscopic endonasal surgery
- Transorbital surgery
- Endoscopic/exoscopic port surgery



Spine

Craniocervical, cervical, thoracic, lumbar, sacral spine

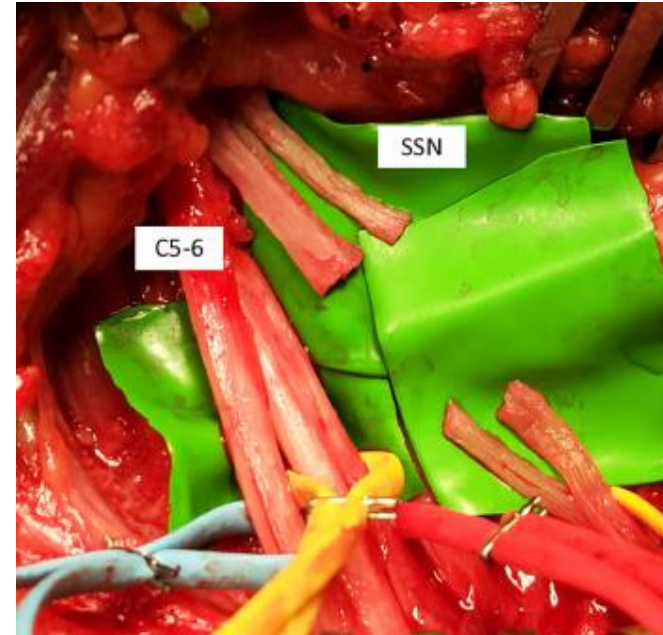
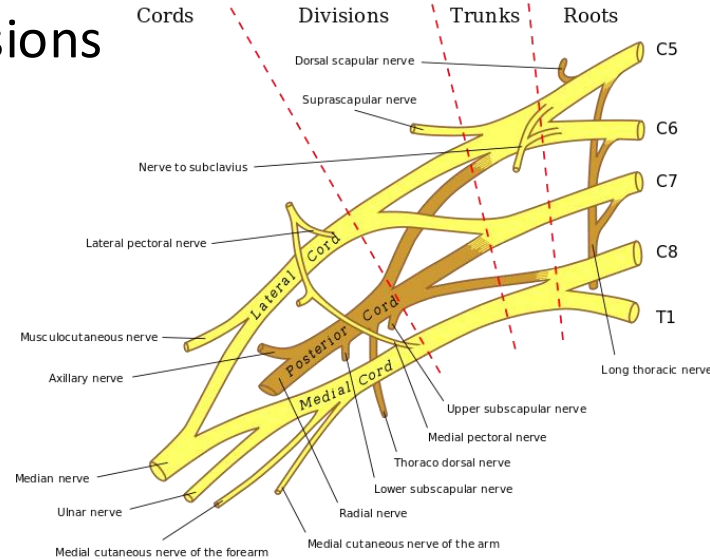
- Degenerative
- Trauma
- Congenital
- Tumor
- Infection/Inflammatory



Peripheral Nerve

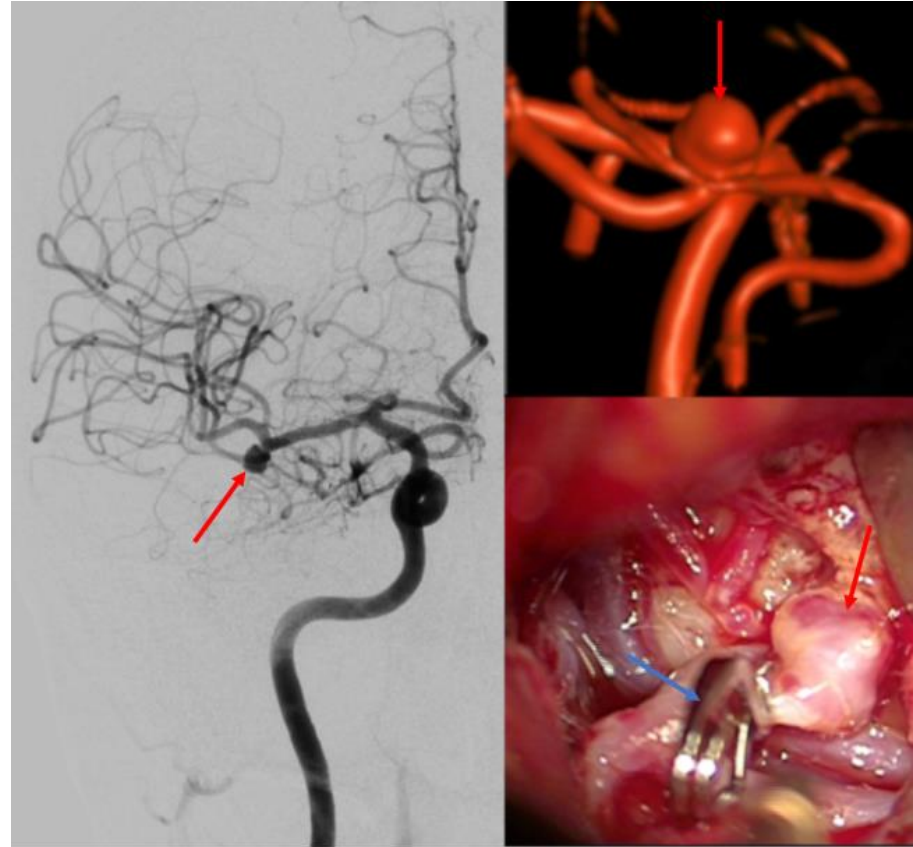
Nerves outside the brain/spinal cord including brachial plexus

- Brachial plexus injuries
- Metabolic and other neuropathies
- Compression syndromes
- Inflammatory lesions
- Tumors
- Pain



Vascular - Open

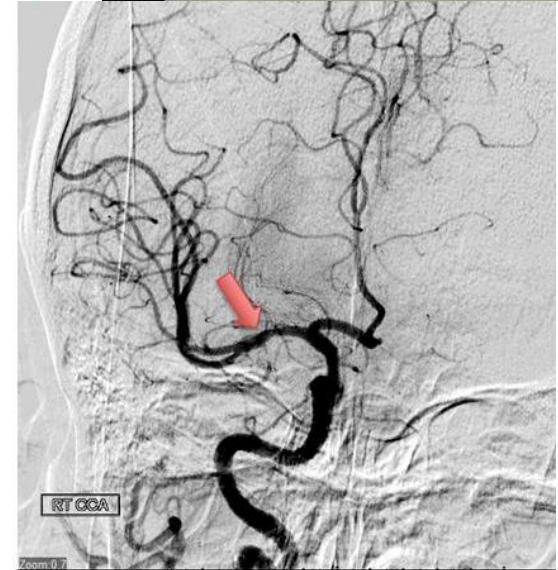
- Aneurysms
- Arteriovenous malformations
- Cavernous malformations
- Fistulas
- Carotid stenosis
- Developmental



Vascular - Endovascular

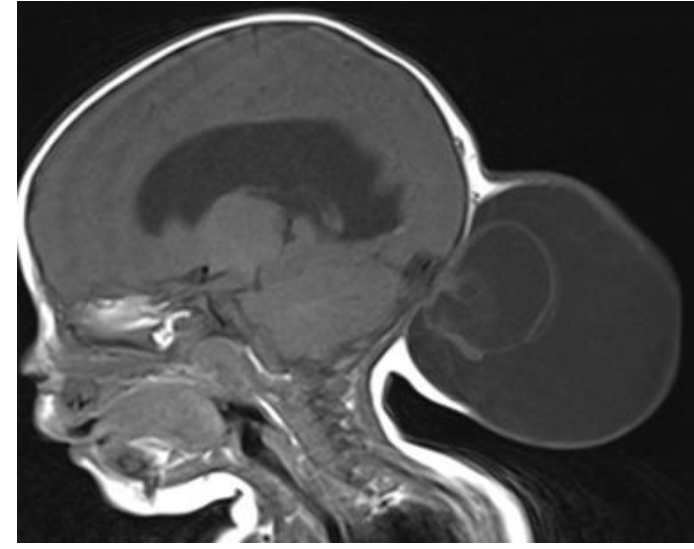
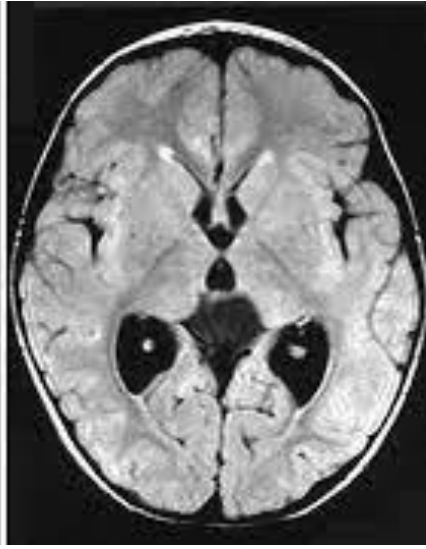
Minimally invasive, access through peripheral arteries

- Aneurysms
- Arteriovenous malformations
- Cavernous malformations
- Fistulas
- Carotid stenosis
- Developmental
- **Stroke**



Pediatrics

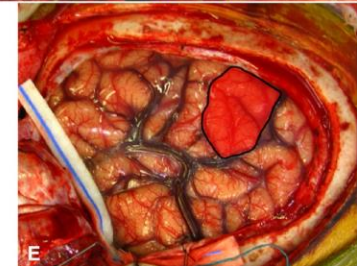
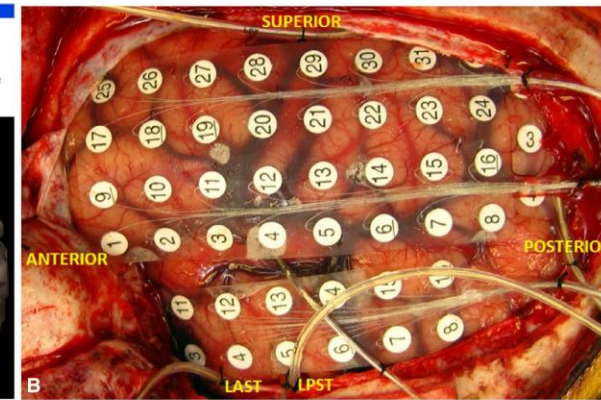
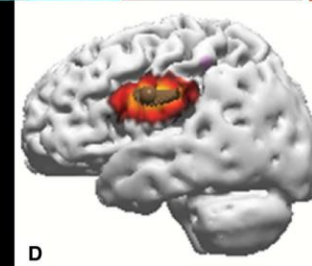
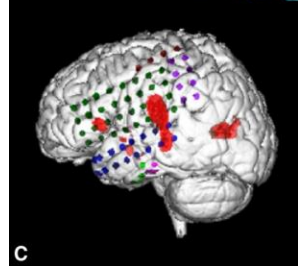
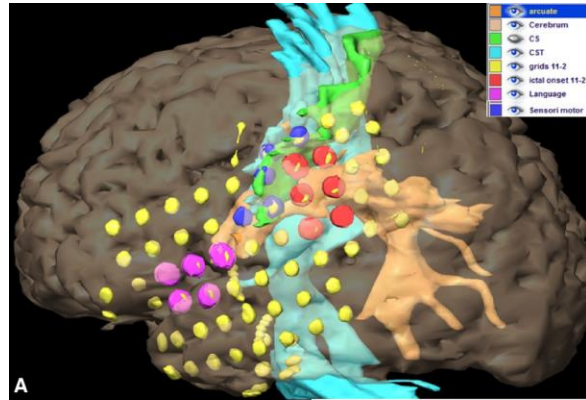
- Congenital/Developmental
- Tumor
- Trauma
- Vascular
- Spine
- Functional
- Hydrocephalus
- **Everything**



Functional

Emphasis on restoring quality of life/neurological function

- Cognitive & neuropsychiatric
- Epilepsy
- Movement disorders
- Pain

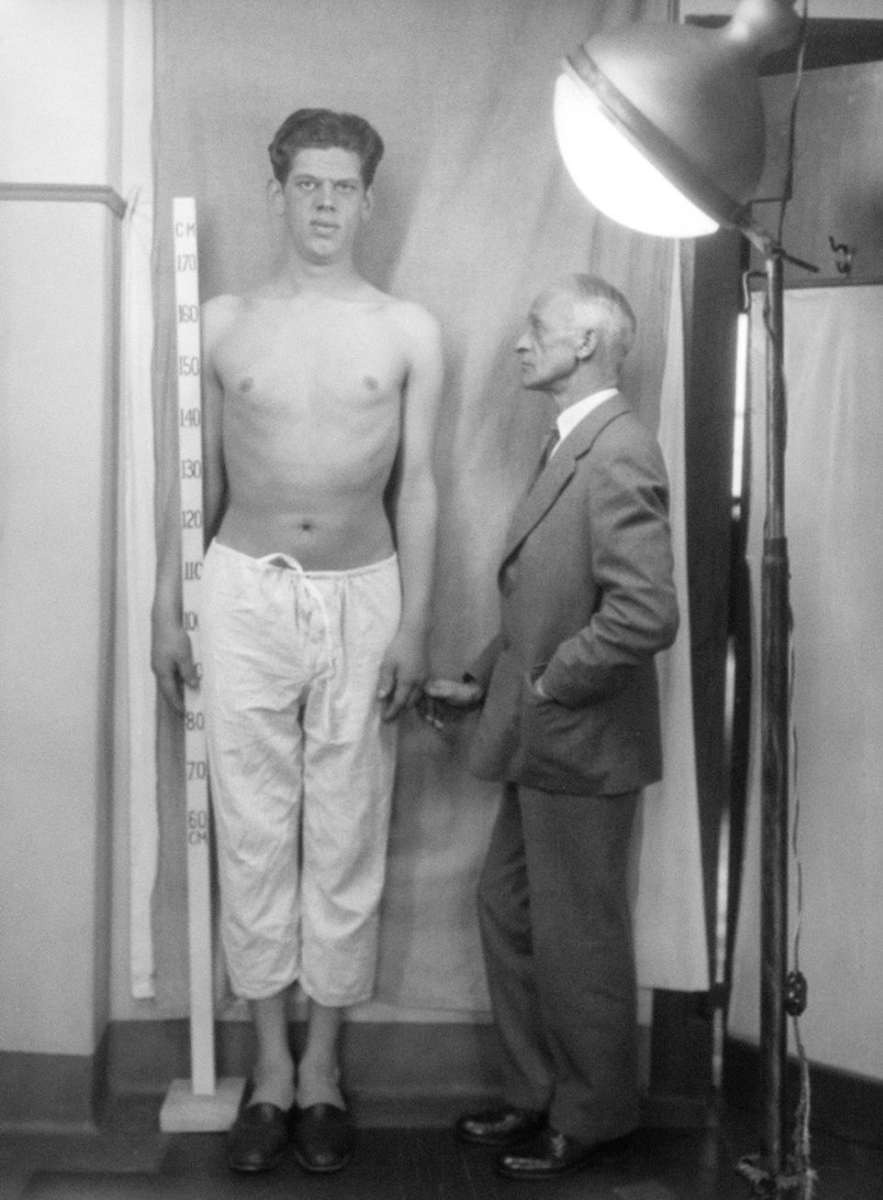


Outline

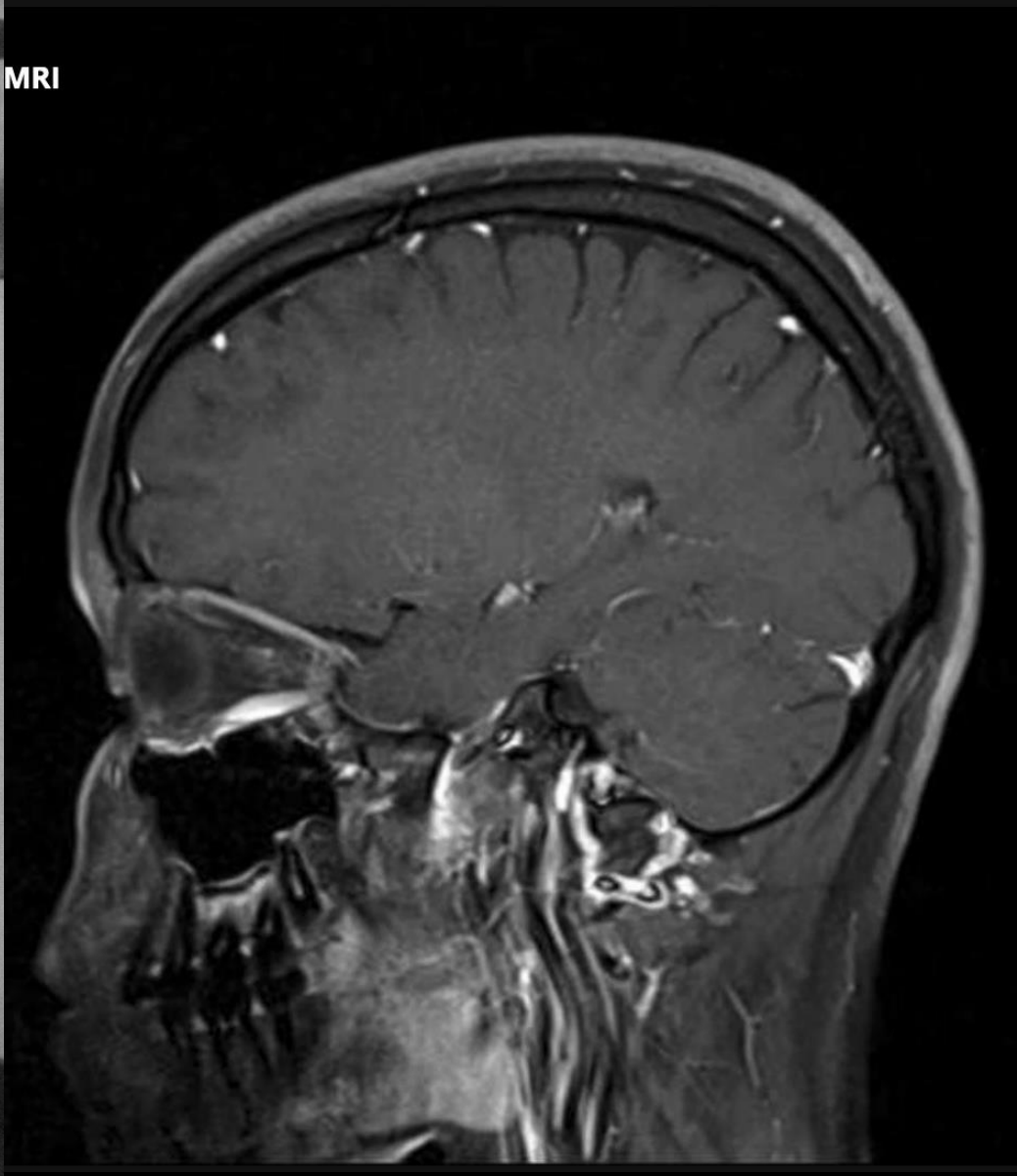
- Neurosurgery: An Overview
- **Computer Integration to Modernize Neurosurgery**
- Robotics in Neurosurgery



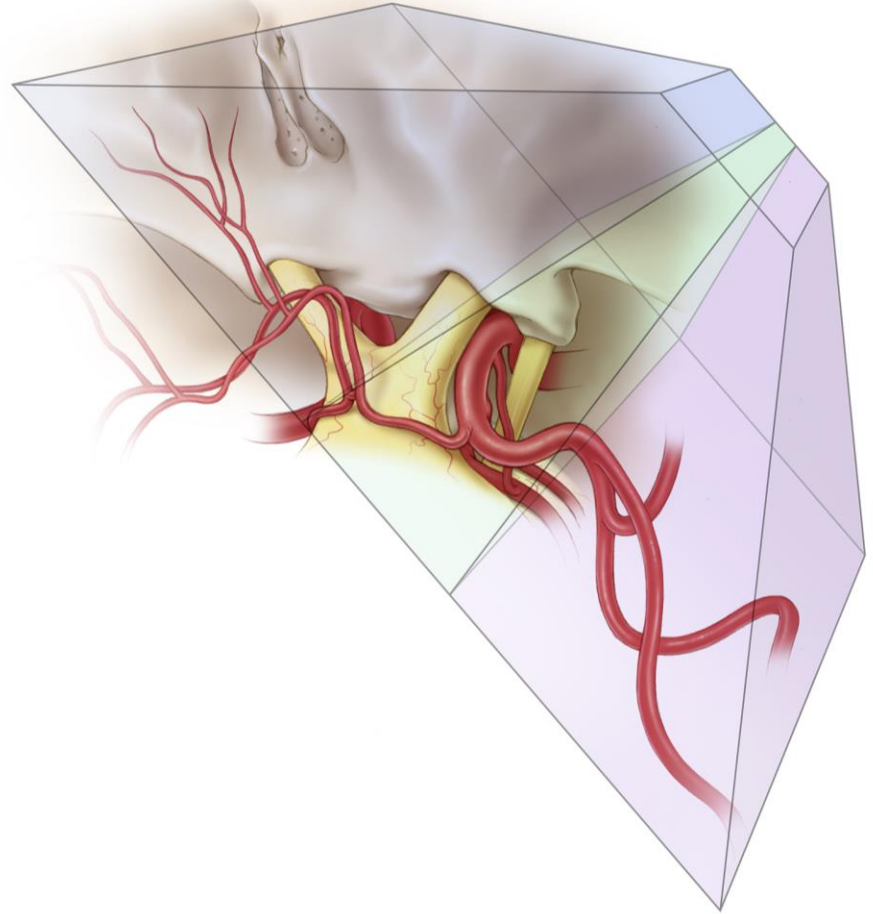
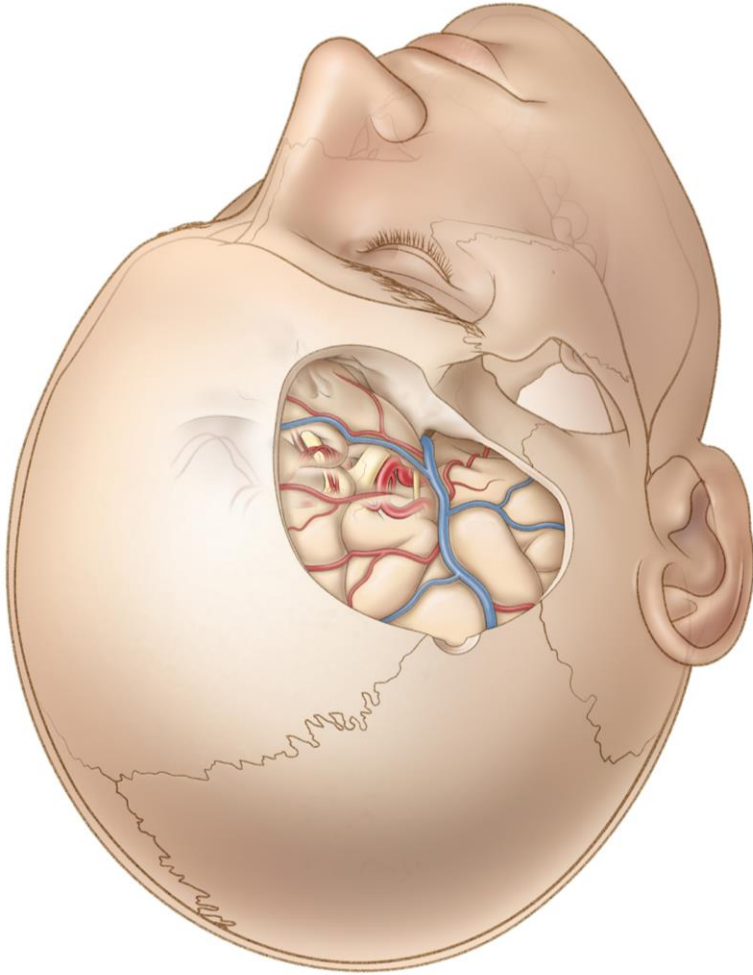
If you needed to get to **center** of head for an operation, how would you get there?



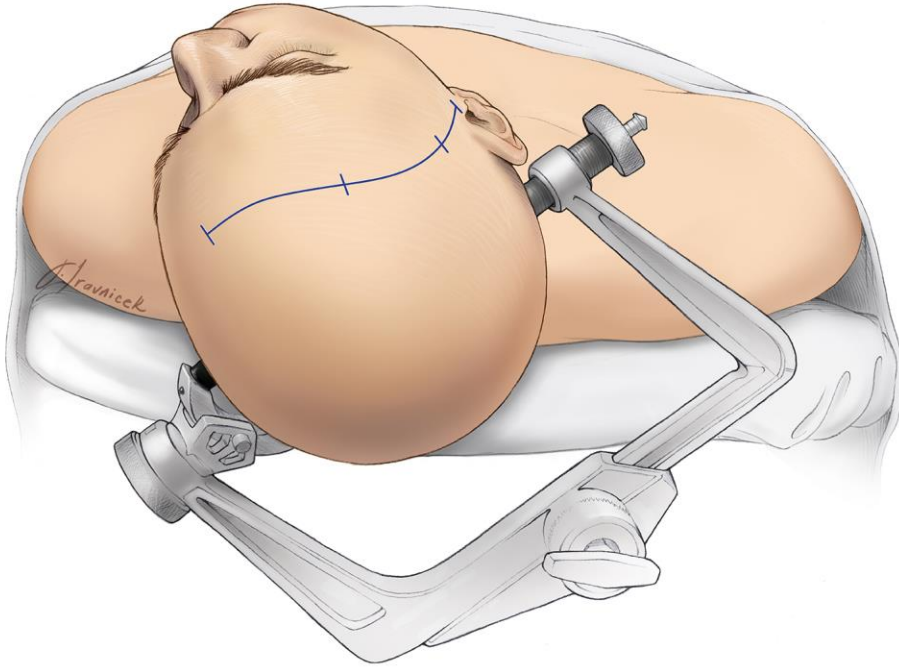
MRI



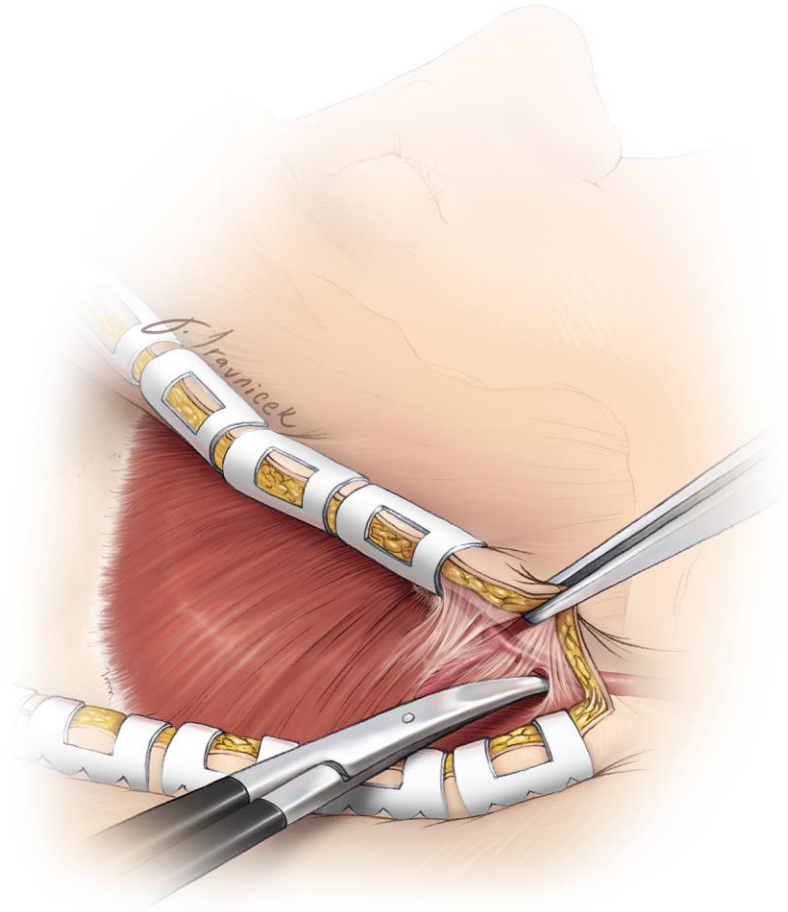
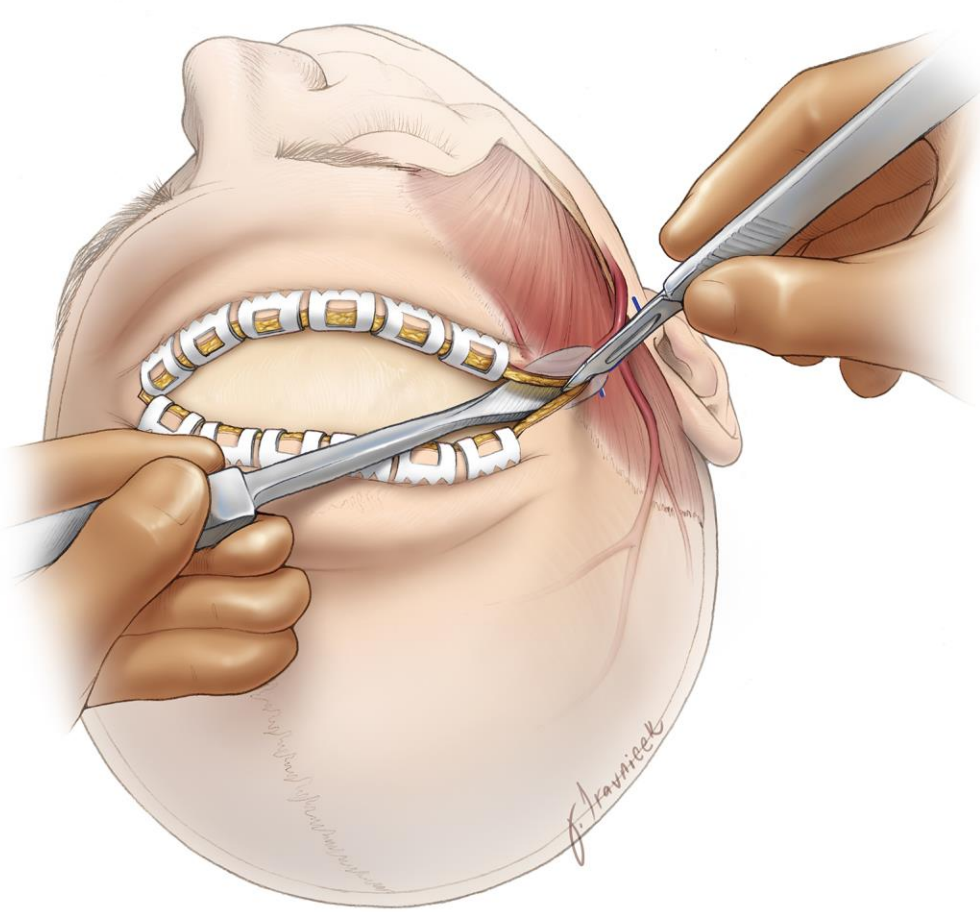
Open Approach: Pterional Craniotomy



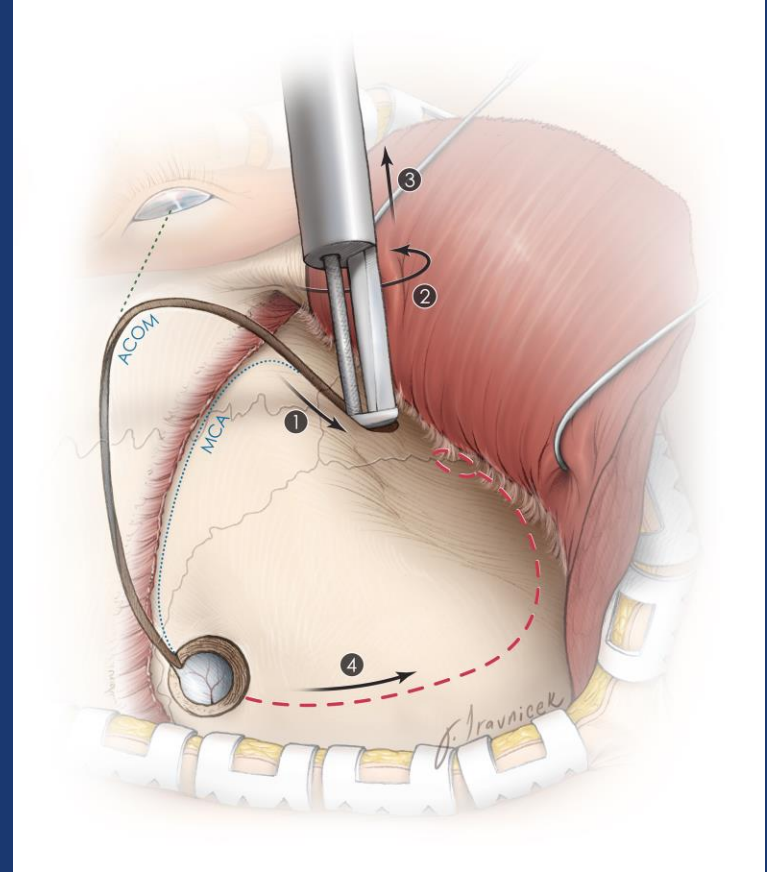
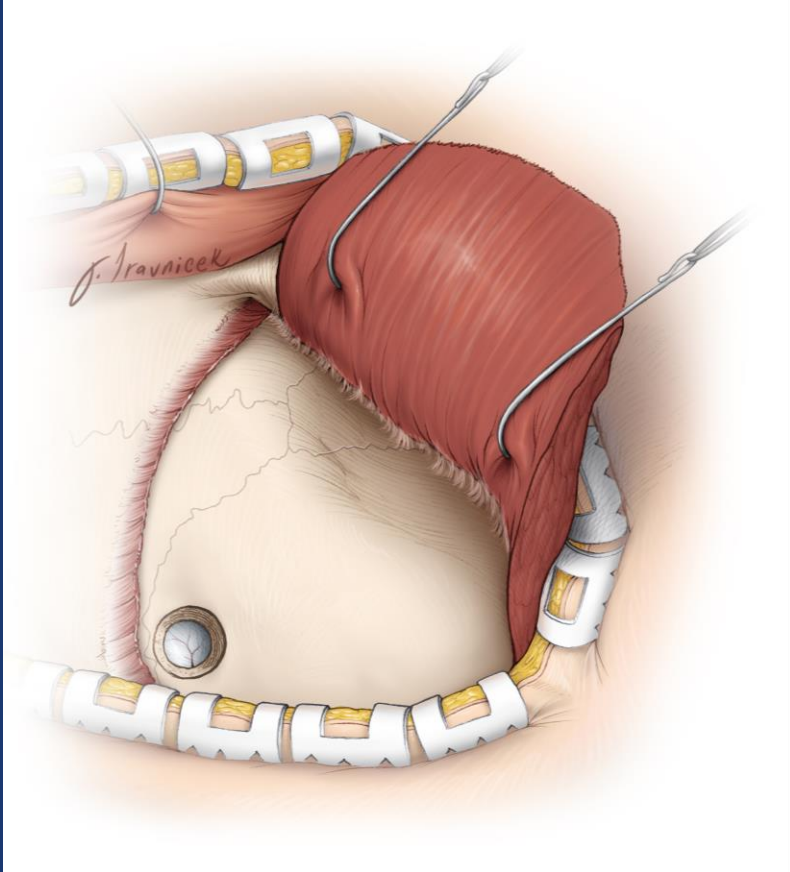
Open Approach: Pterional Craniotomy



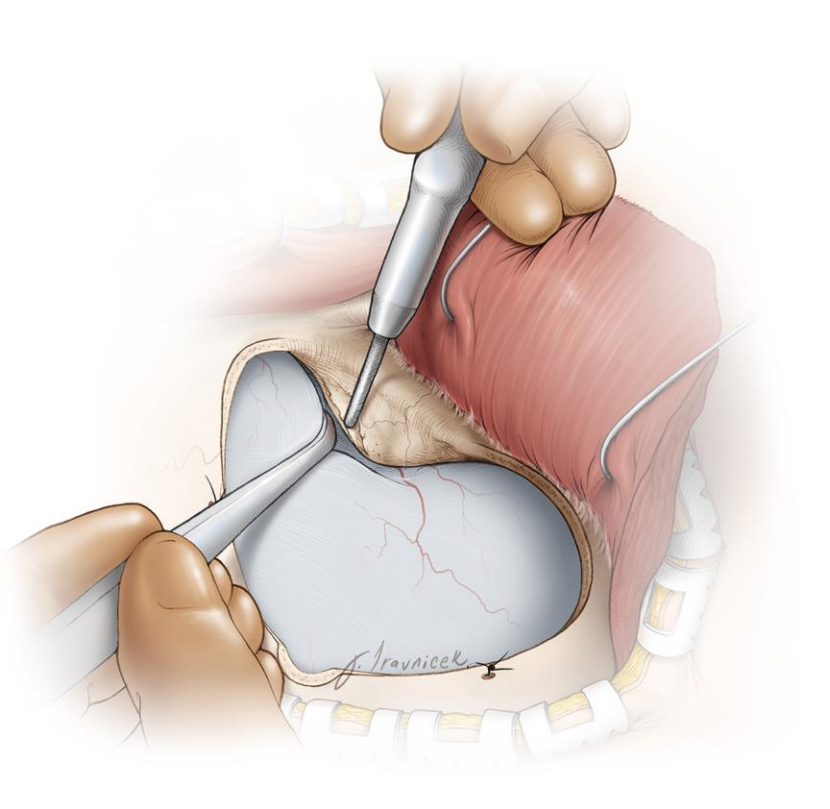
Open Approach: Pterional Craniotomy



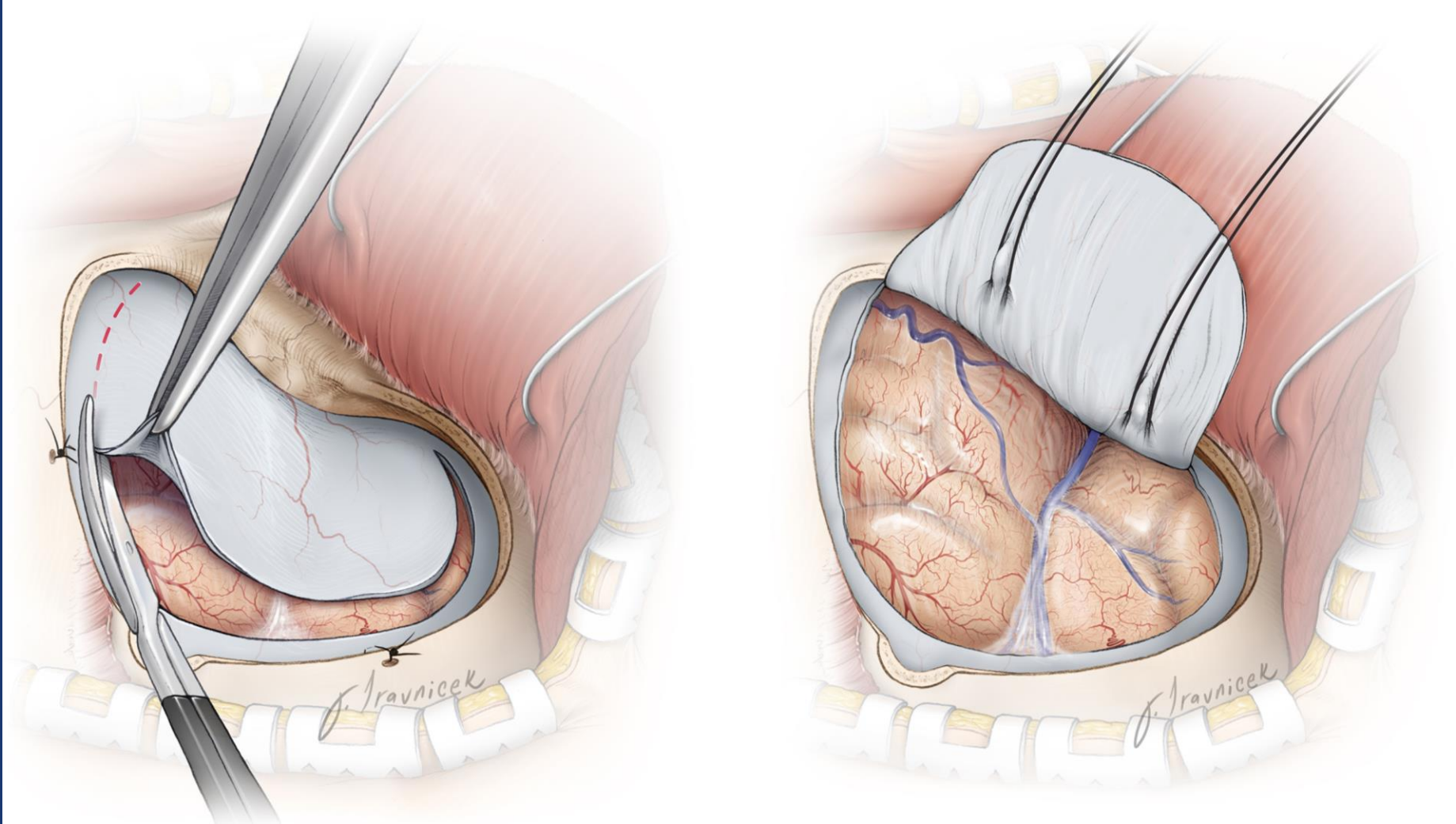
Open Approach: Pterional Craniotomy



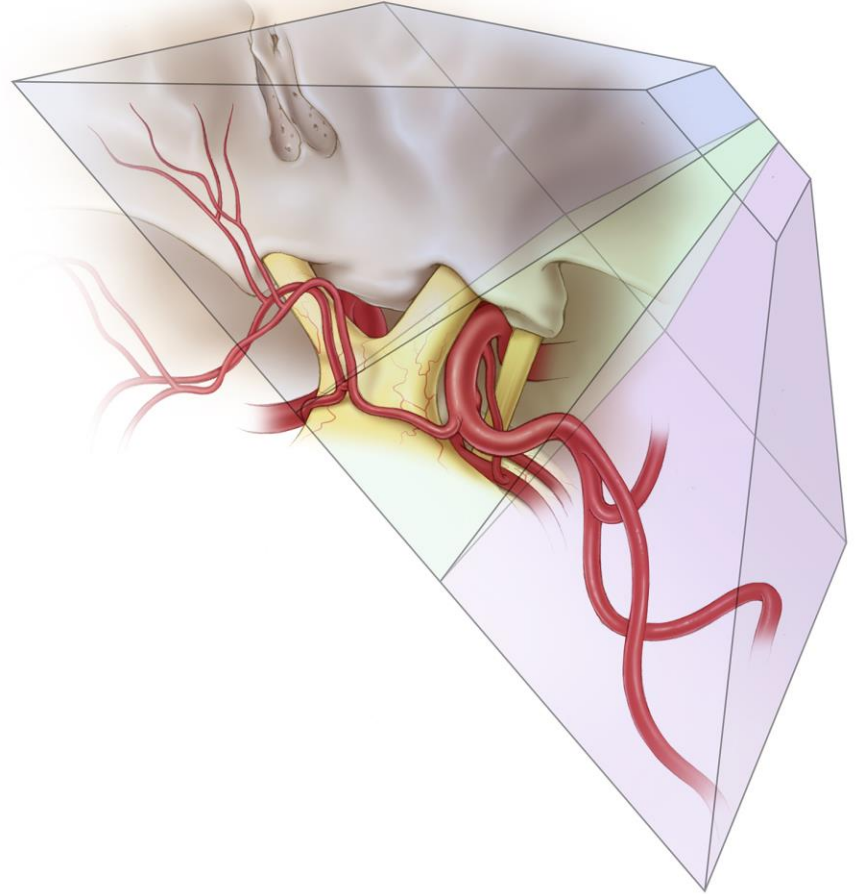
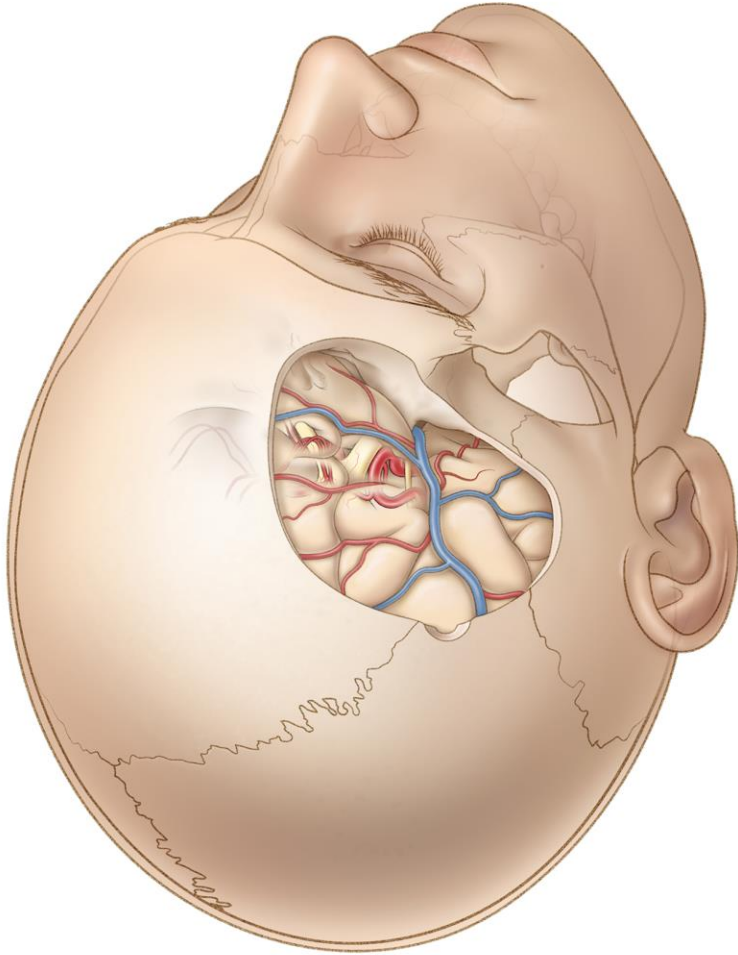
Open Approach: Pterional Craniotomy



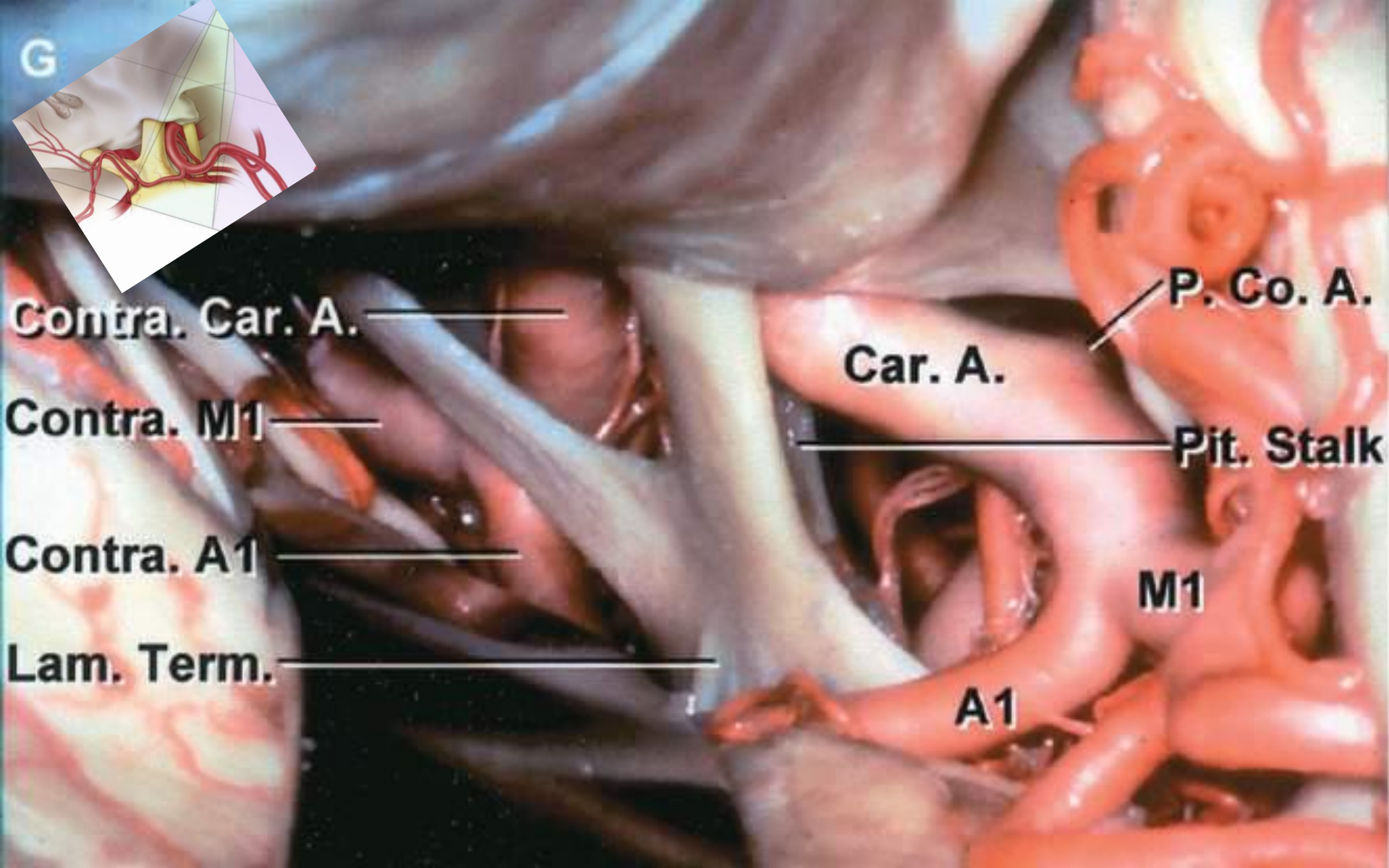
Open Approach: Pterional Craniotomy



Open Approach: Pterional Craniotomy



G



Contra. Car. A.

Contra. M1

Contra. A1

Lam. Term.

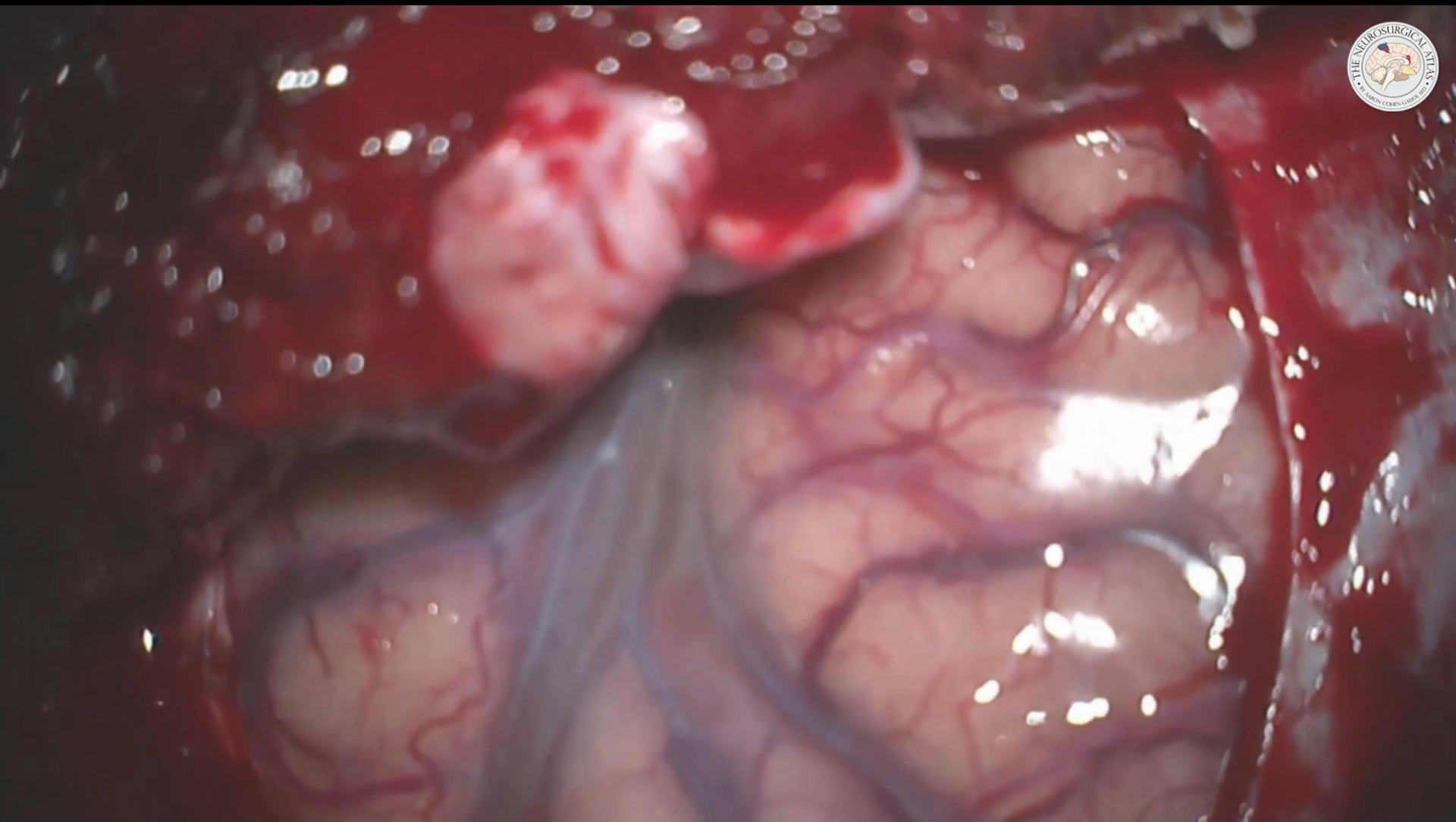
Car. A.

P. Co. A.

Pit. Stalk

M1

A1



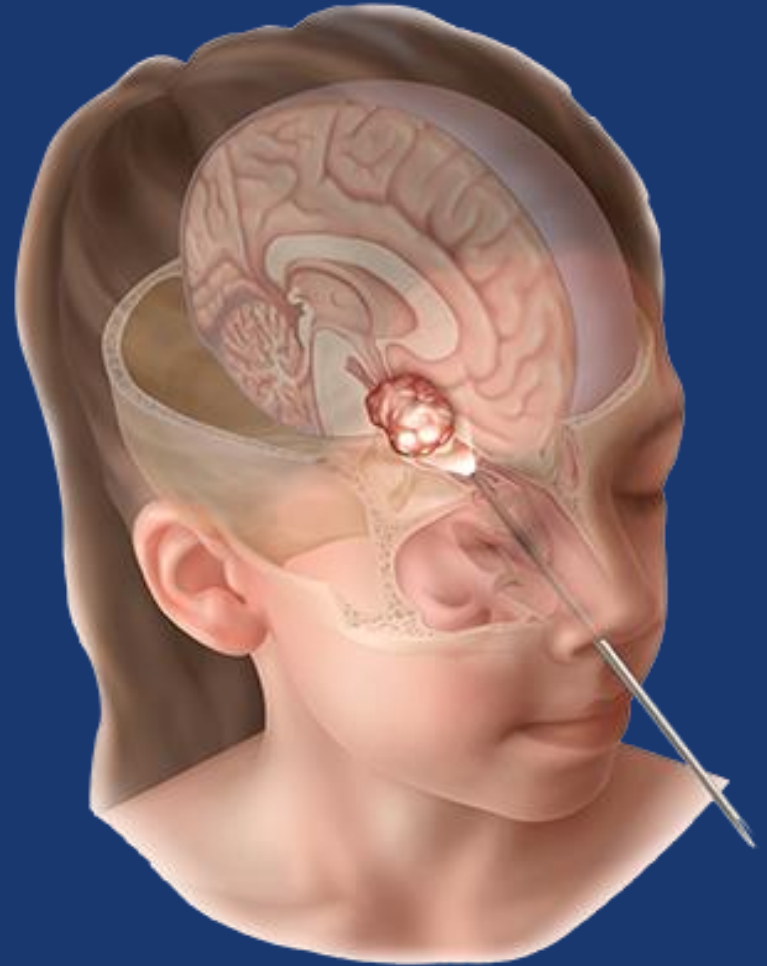
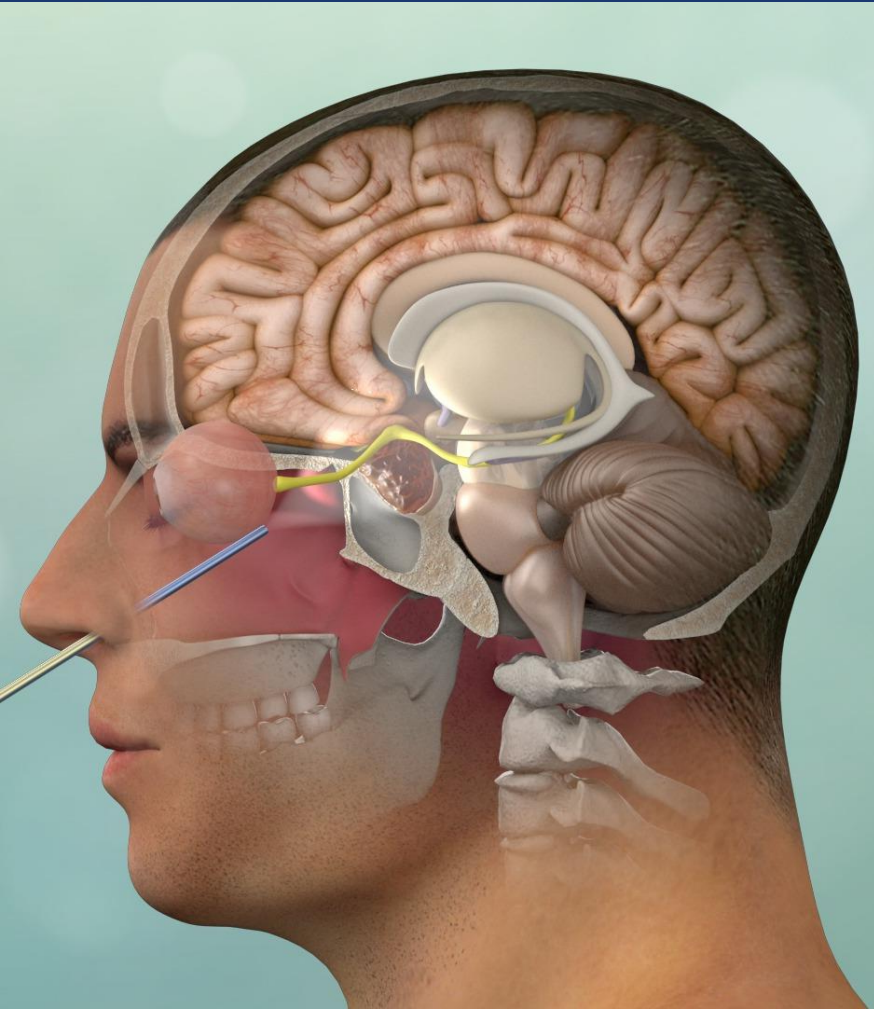
Drawbacks of Open Surgery

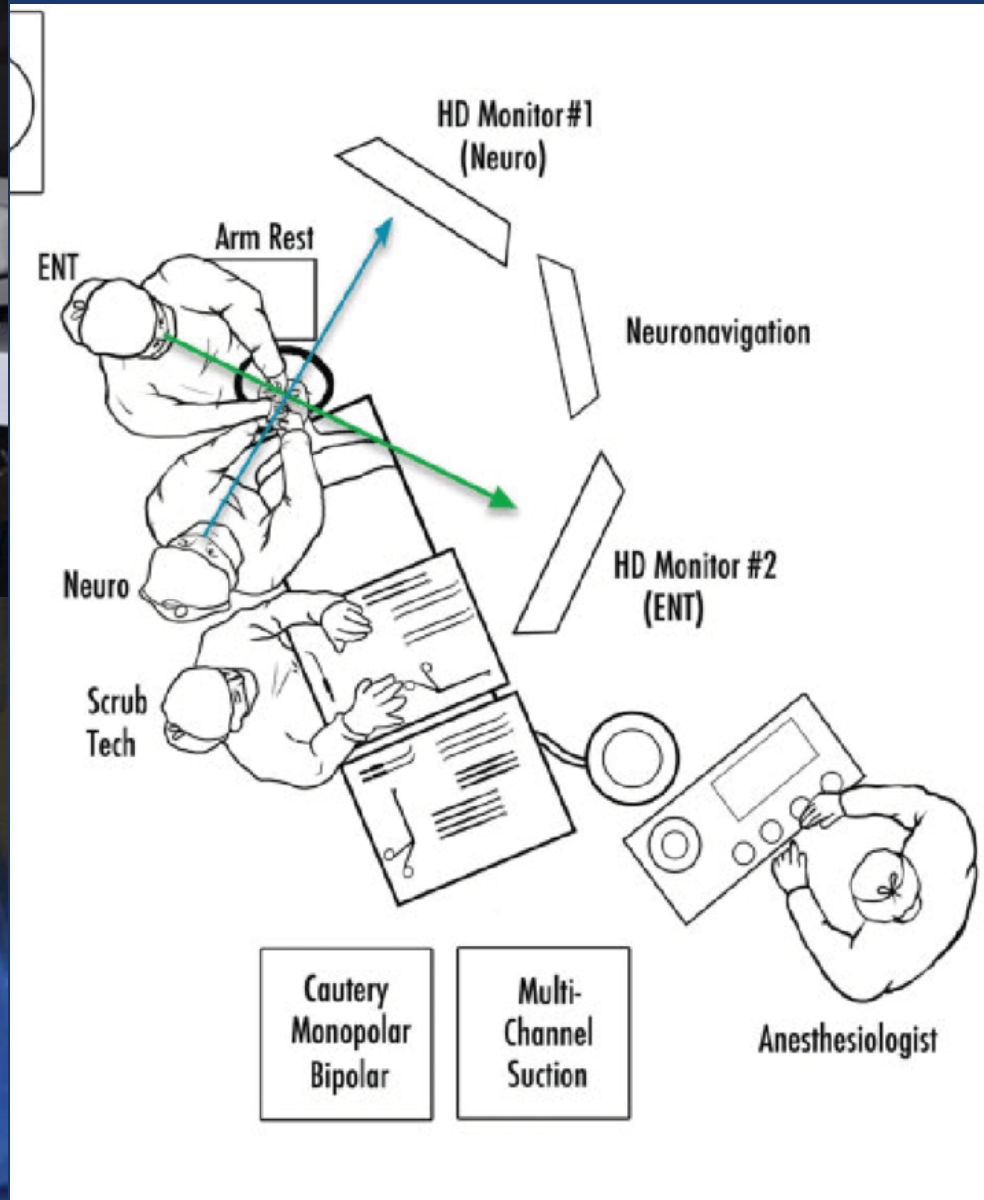
Wound healing, infection, neurological damage, etc

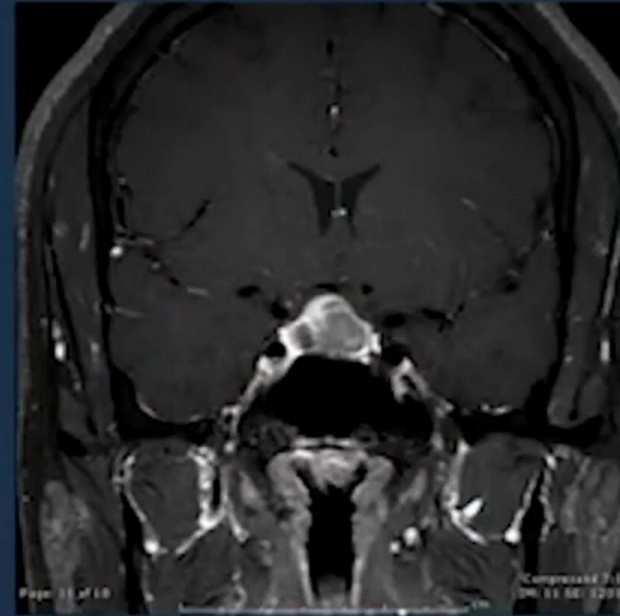
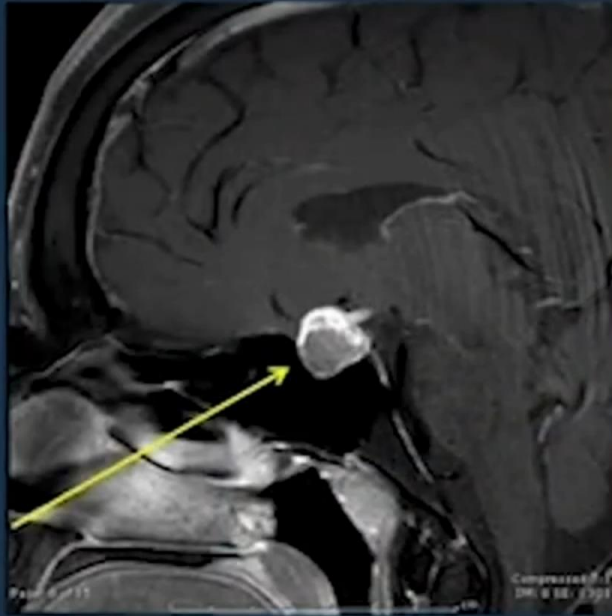


Alternative to an **open** approach?

Endoscopic Endonasal Approach





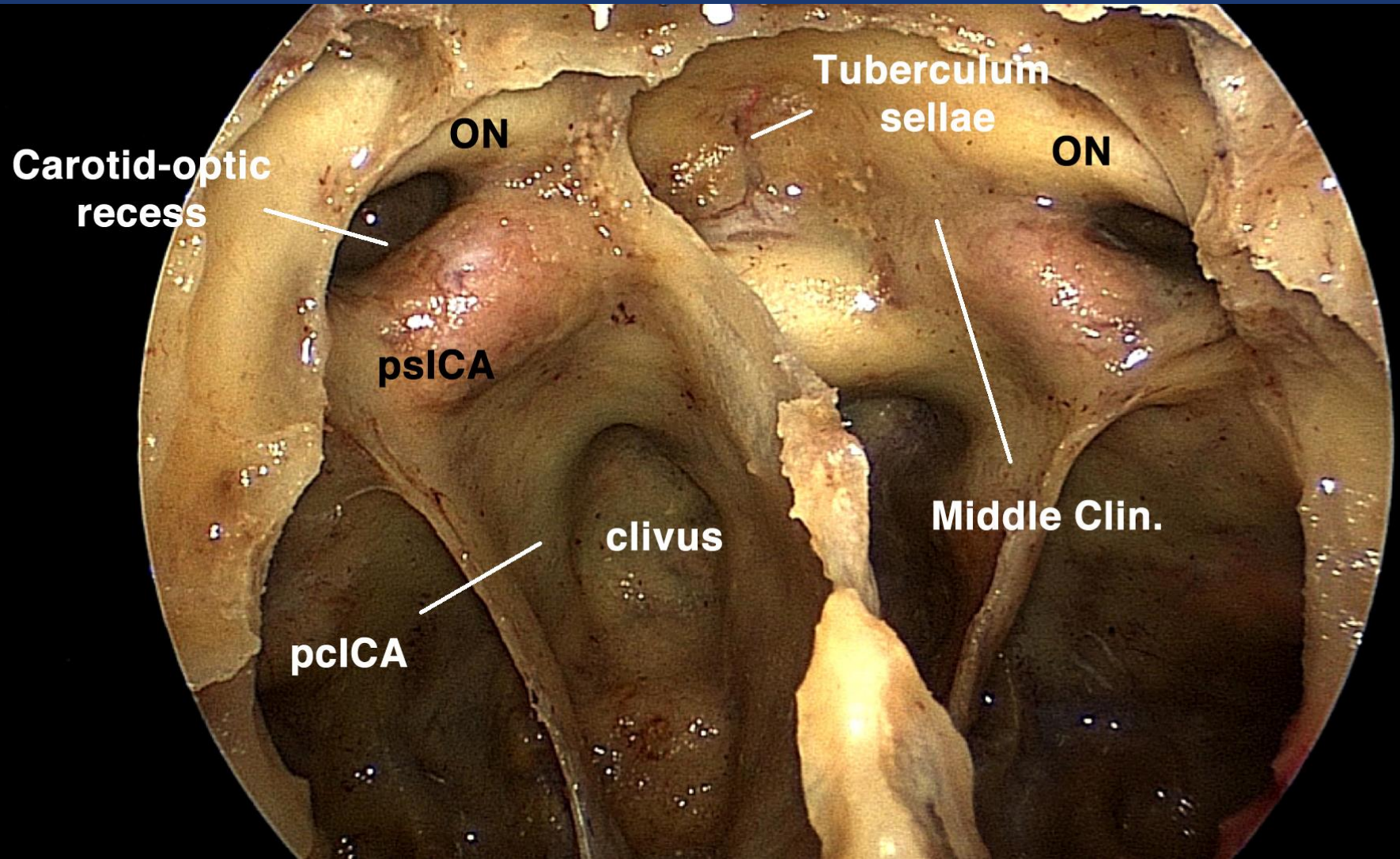


Endonasal endoscopic removal of pituitary macroadenoma

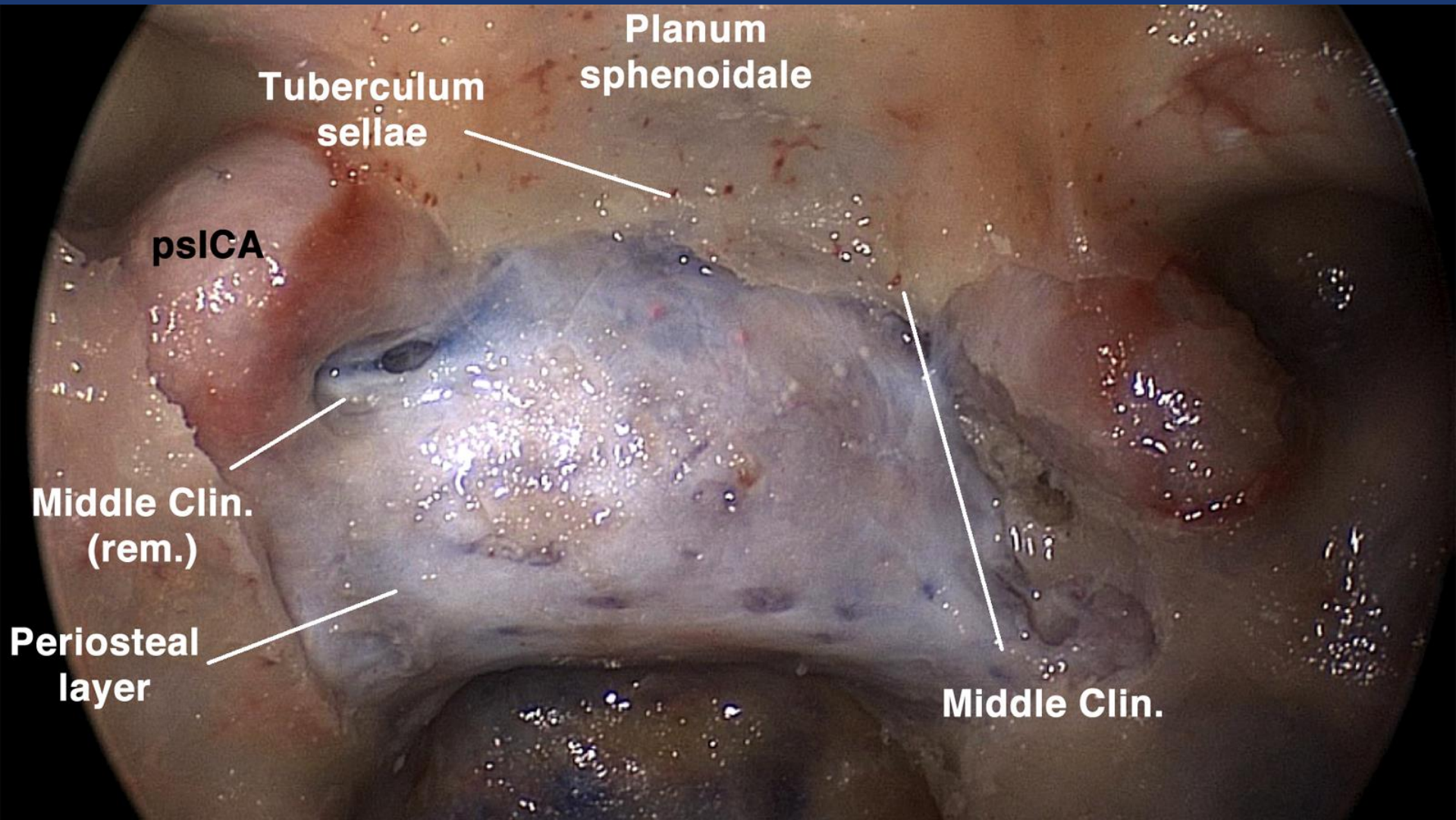
is shown. The initial portion of the
procedure involves removing bone at the

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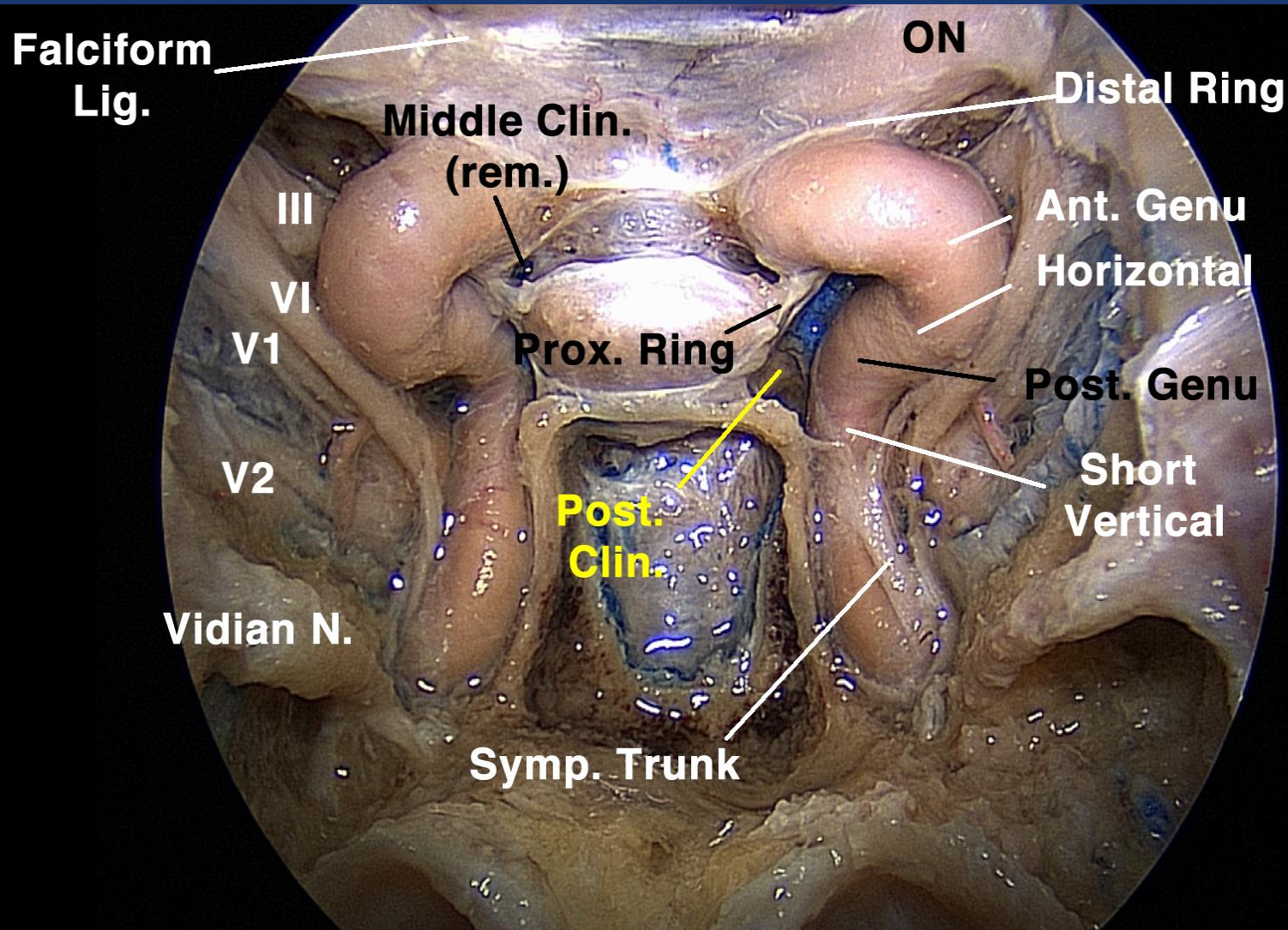
Endoscopic Endonasal Approach

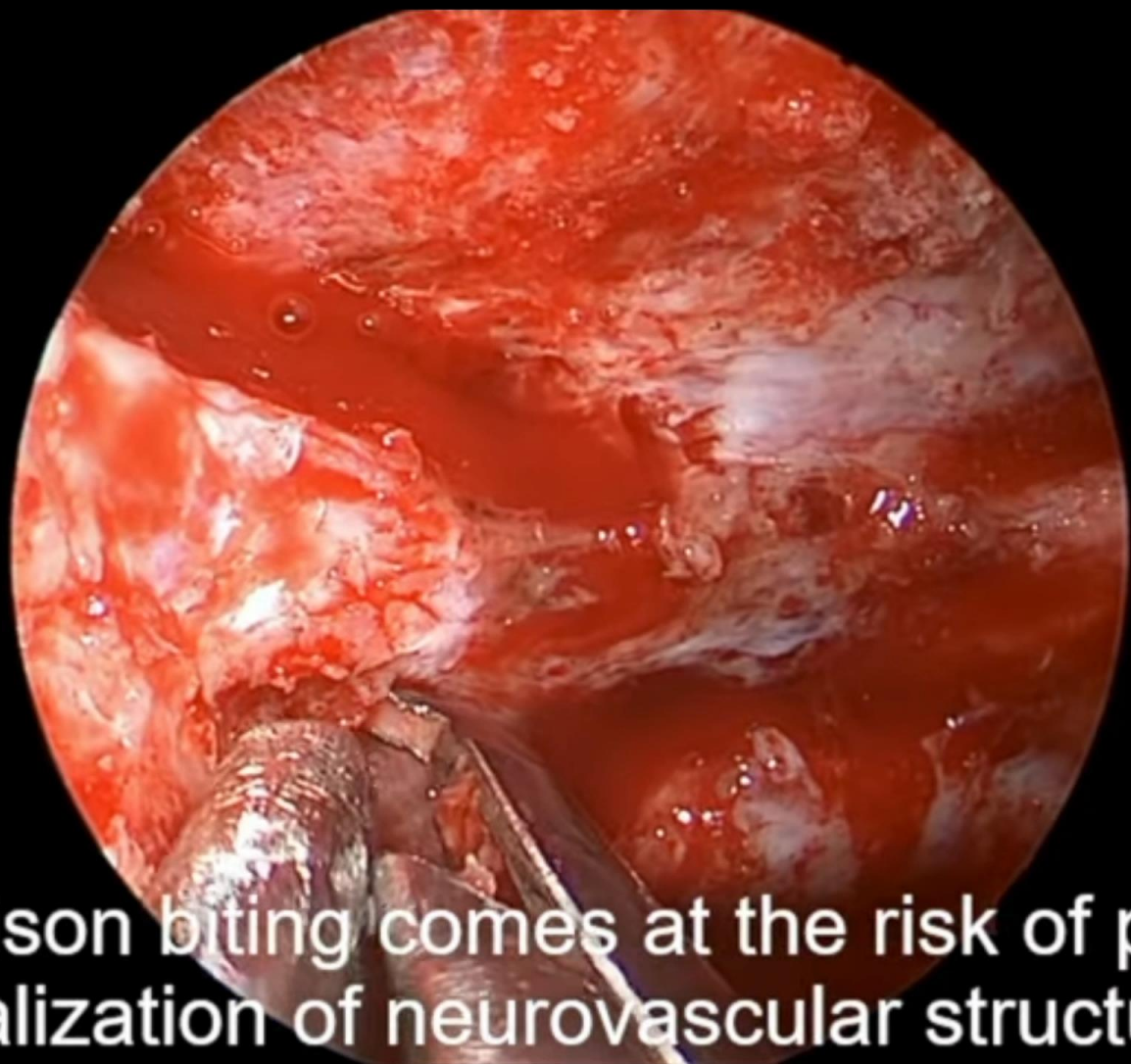


Endoscopic Endonasal Approach



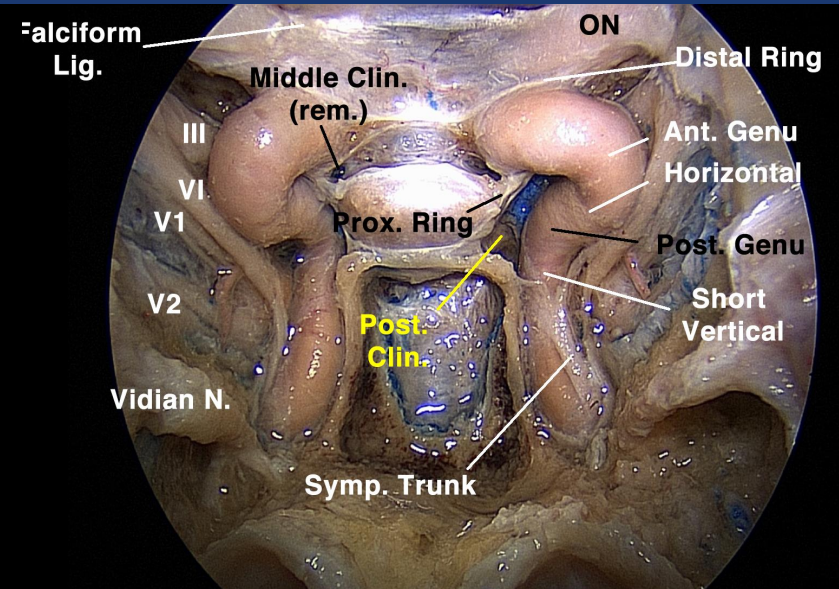
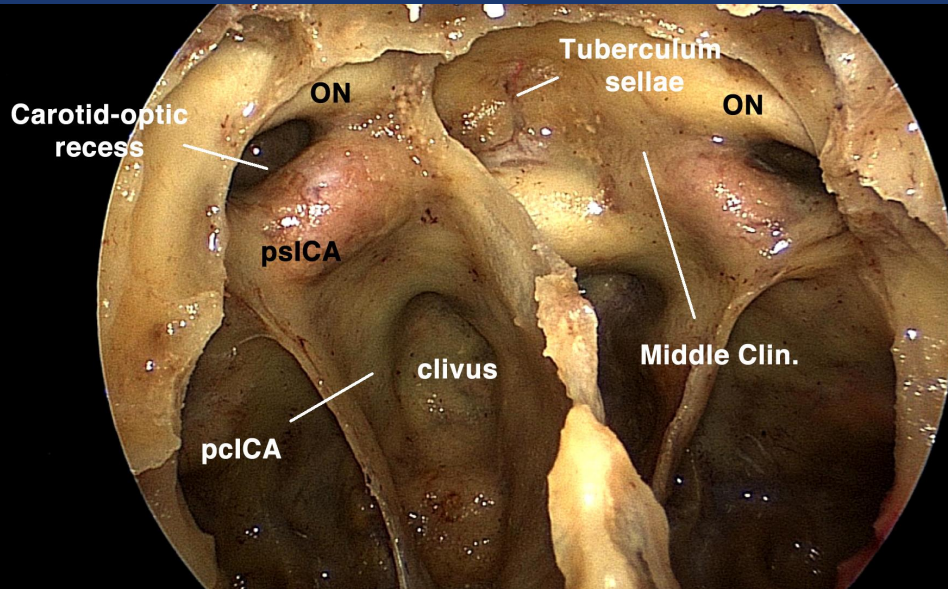
Endoscopic Endonasal Approach



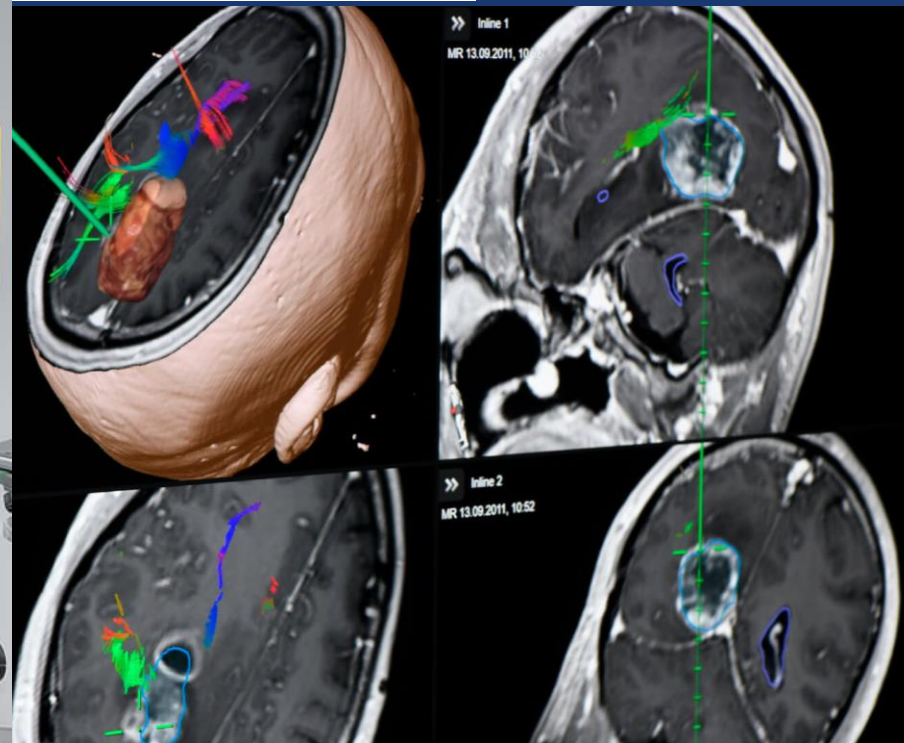
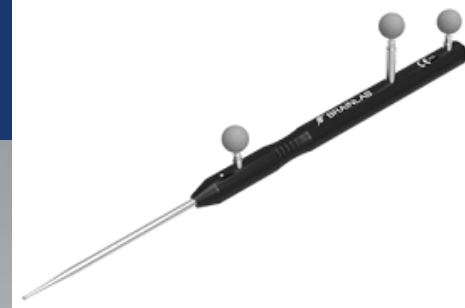


Kerrison biting comes at the risk of poor visualization of neurovascular structures

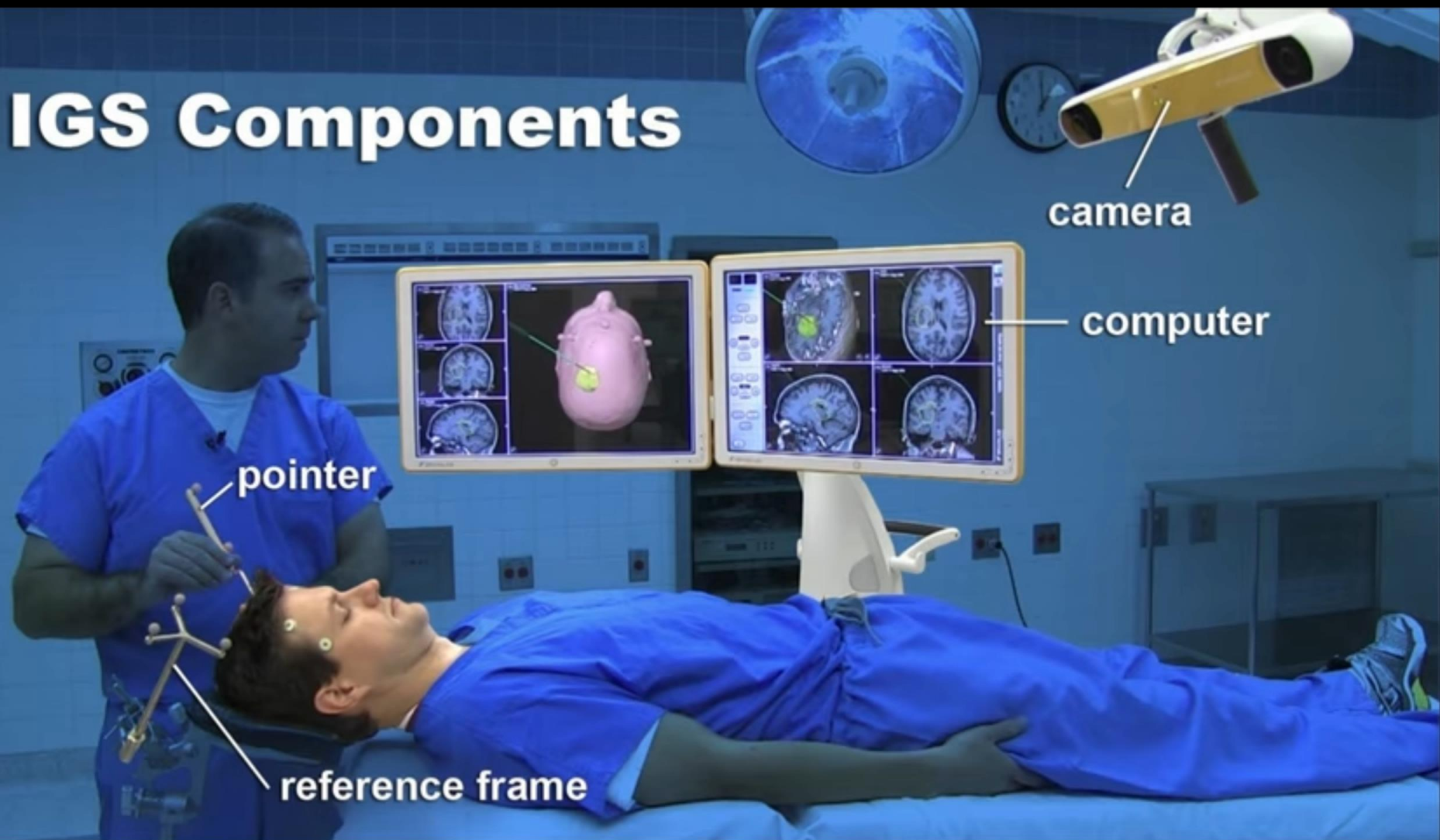
How then is surgery safely performed around such high-stakes anatomy?



Surgical Navigation Systems



IGS Components



Surgical Navigation

Registration defines a correlation between a reference point in a 3D data set such as CT or MRI with the corresponding reference point in a patient.

Most navigation systems achieve **position errors on the order of 2mm**

- Vulnerable to **physical displacement** or **computer malfunction**
- Requires **repeated visual confirmation** of registration accuracy during surgery



Surgical Navigation

Surgical navigation systems display the **same image information even as anatomy changes**.

- Relationship between endoscopic view and navigation view is **lost** over time

Intra-operative cone-beam or CT imaging is a way to **update** visualization

- BrainLab Brainsuite iCT
- Medtronic O-Arm system



Drawbacks of Intra-Operative CT

- Additional radiation, operative time, and costs.
- Inferior reconstruction quality if using cone-beam.



Rationale for improving navigation

- Enhance **patient safety** and **outcomes** by reducing potential **complications** and **radiation exposure**
- Reduce costs by improving **clinical workflow** and clarity of **intraoperative visualization**



How then do we **improve**
navigation during **endoscopic**
endonasal surgery?

Proposal:

Utilize images from the **endoscope** as a basis for **registration to pre-operative imaging** and **reconstruction** of anatomical surfaces.

Quantitative Endoscopy (QE)

Goal: transform the endoscope from a visualization device to an instrument for quantitative 3D measurement.

Endoscopic measurements combined with CT or MRI to provide:

- enhanced navigation (**goal accuracy 0.5mm**),
- tissue surface reconstruction,
- and fused image visualization.



Video-Based Navigation System Overview

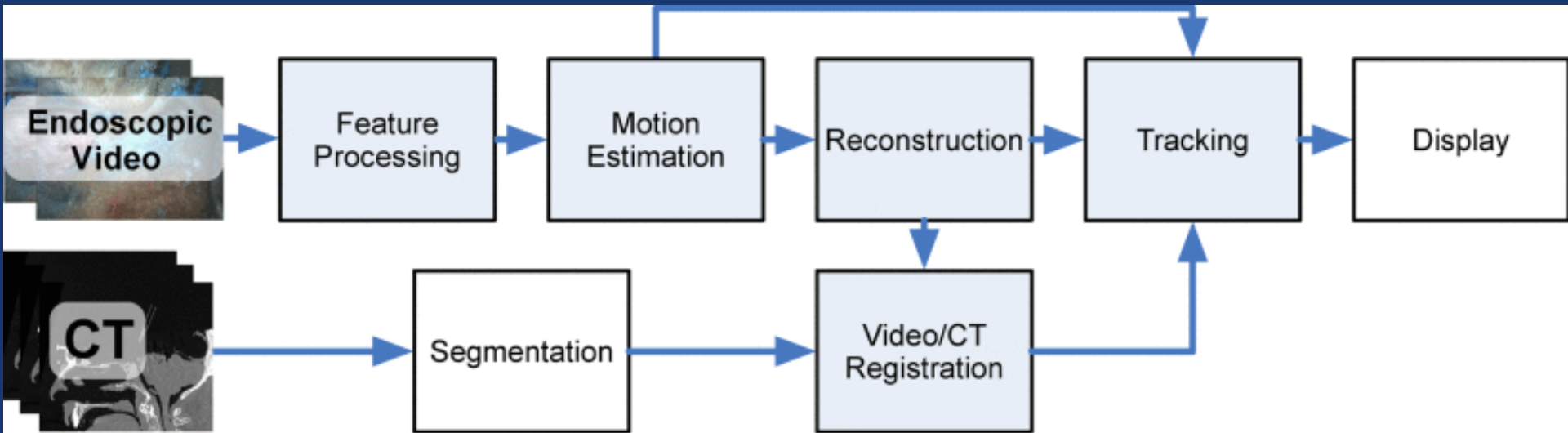
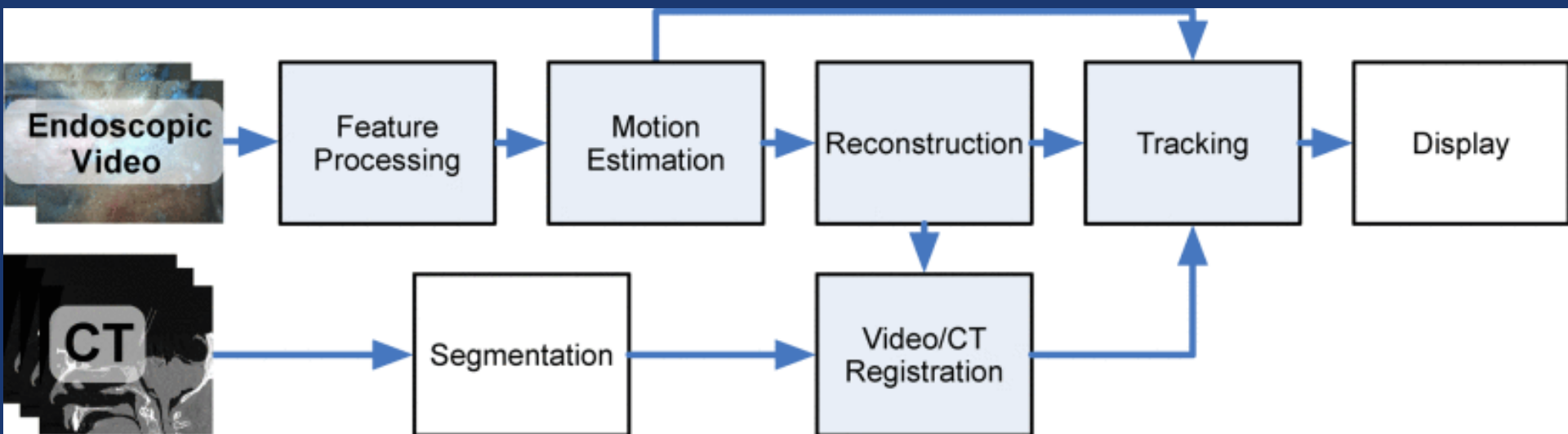


Image features detected and matched in two temporally adjacent images.

These matching pairs are then used to estimate the camera motion using a robust estimator we have developed

Video-Based Navigation System Overview



Once the camera motion is estimated, the 3D location of the matched features are reconstructed.

The reconstructed 3D surface points are then passed to the 3D-3D registration component.

Target Registration Error (TRE)

TRE_1	Metric for evaluating pointer-based methods
TRE_2	Metric for evaluating tracker-based and video-based methods
NGE	Same as TRE_2 , however, the target is not visible in the endoscope image.

$$TRE_1 = \left\| \mathbf{p}_{CT} - \left({}^{CT}T_{Navigation} \right) \mathbf{p}_{pointer} \right\|$$

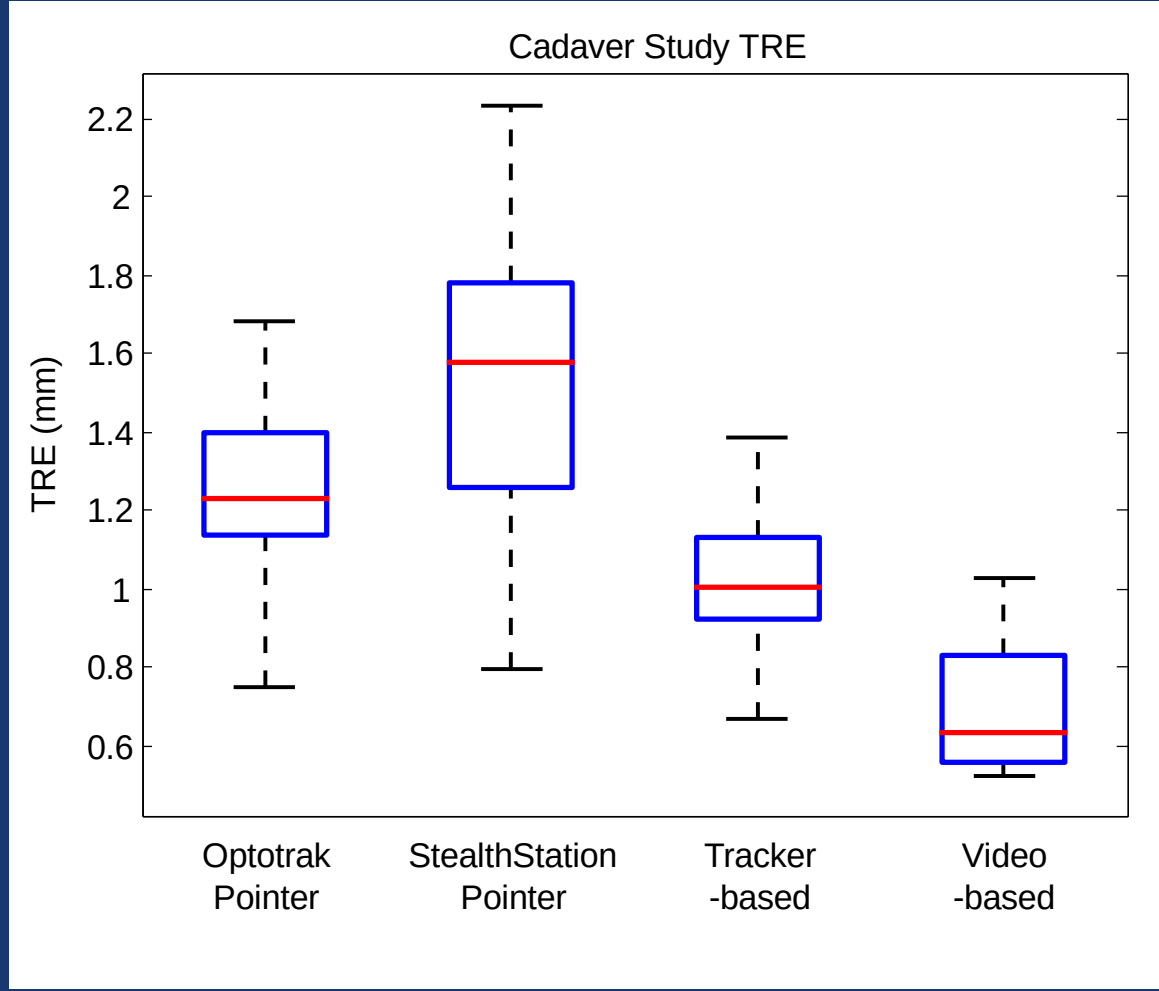
$$TRE_2 = \left\| \mathbf{p}_{CT} - \left(\mathbf{t} + \mathbf{r} \left(\frac{\mathbf{r} \cdot (\mathbf{p}_{CT} - \mathbf{t})}{\mathbf{r} \cdot \mathbf{r}} \right) \right) \right\| \quad \text{where} \quad \mathbf{r} = RK^{-1} \mathbf{q}_{image} - \mathbf{t}, \quad \text{experir}$$



Sinus Reconstruction



Key result: TREs using video-CT methods are measurably improved over traditional methods



Quantitative Endoscopy (QE)

Incorporation of computational vision algorithms with traditional navigation methods provides several benefits.

- Improves usability of **existing** navigation technology in sinus surgery with **no additional cost or equipment**.
- **Minimal disruption** to the surgical workflow.



Outline

- Neurosurgery: An Overview
- Computer Integration to Modernize Neurosurgery
- Robotics in Neurosurgery



Historical Perspective - PUMA



- First reported use was the PUMA system in 1985 for stereotactic, CT-guided brain biopsy.
- Quickly abandoned over safety issues



Historical Perspective - Neuromate



- Neuromate for brain electrode implantation
- Neuromate is still available on the market

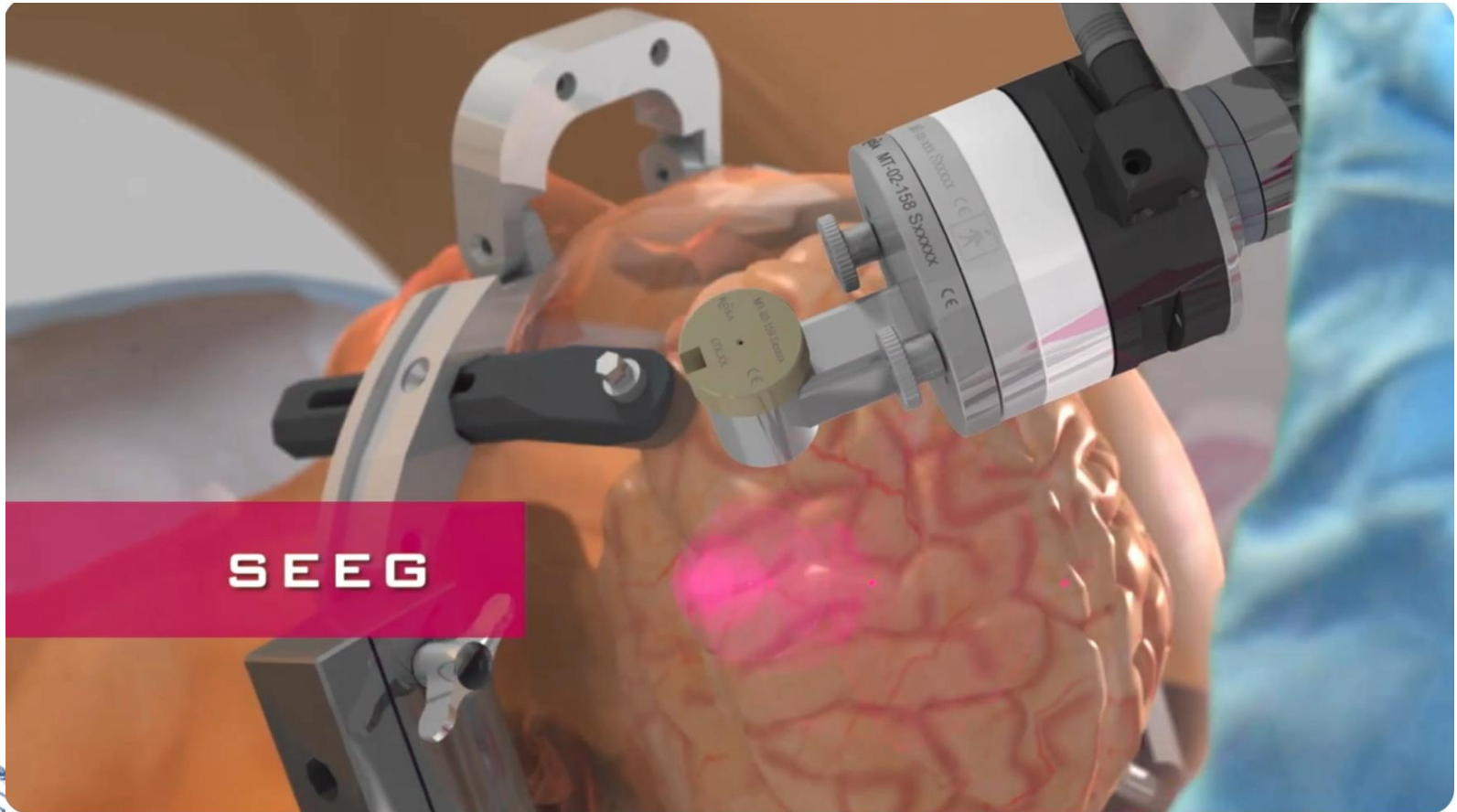


Stereoelectroencephalography

- aka sEEG
- Recording of brain activity to determine sources of seizures
- Robotics have increased the accuracy and safety
- Robots don't suffer from fatigue or tremor

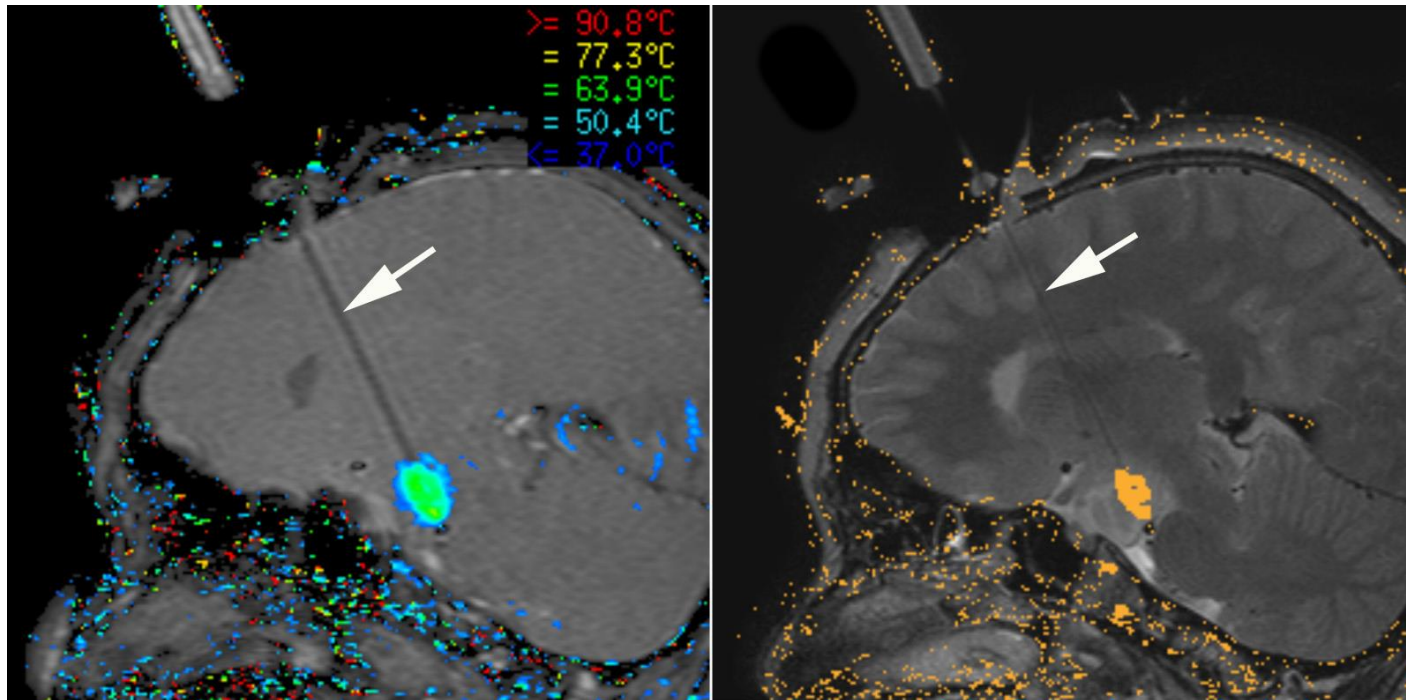


Stereoelectroencephalography



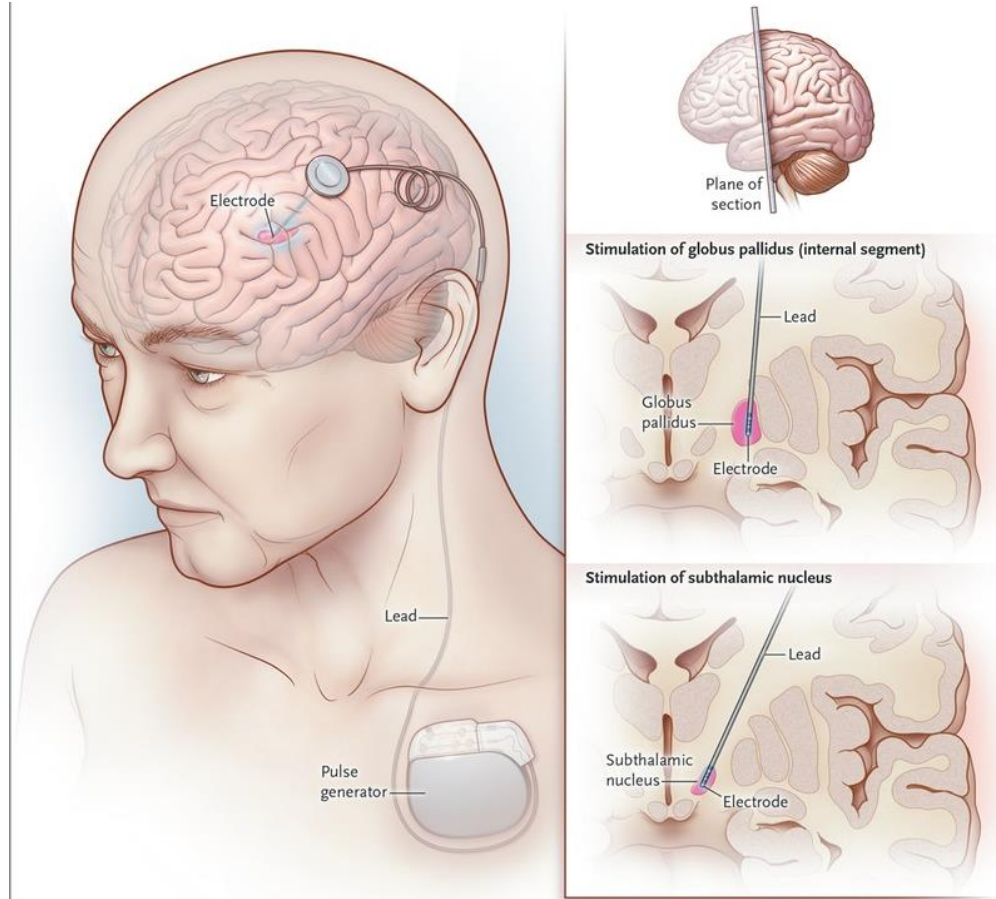
RF-THC and LiTT

- Radio-Frequency Thermocoagulation and Laser Interstitial Thermal Therapy (LiTT)
- Used for drug-resistant epilepsy
- Enables ablation before removing the SEEG electrodes



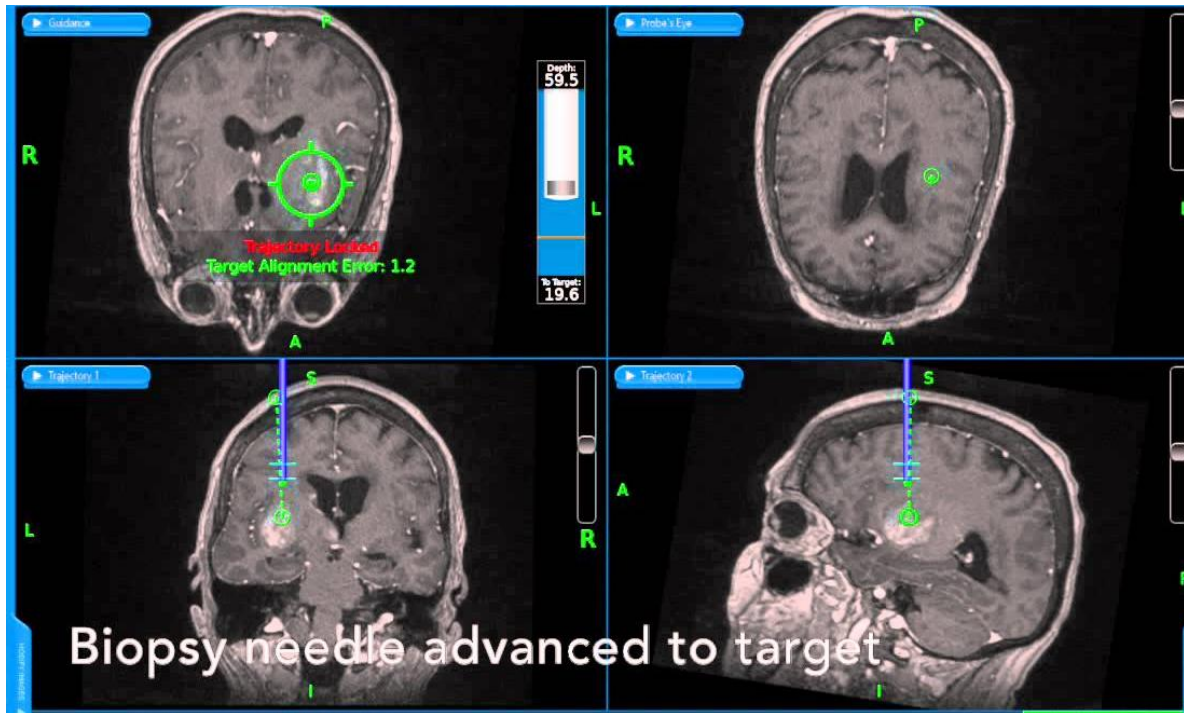
Deep Brain Stimulation

- Used to treat movement disorders and drug-resistant epilepsy.
 - e.g. Parkinson's disease



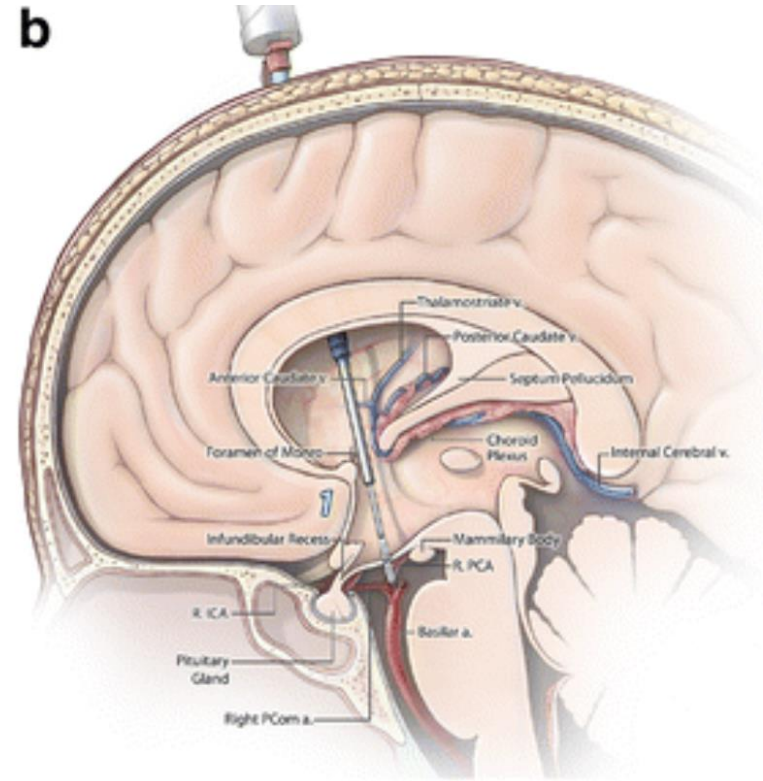
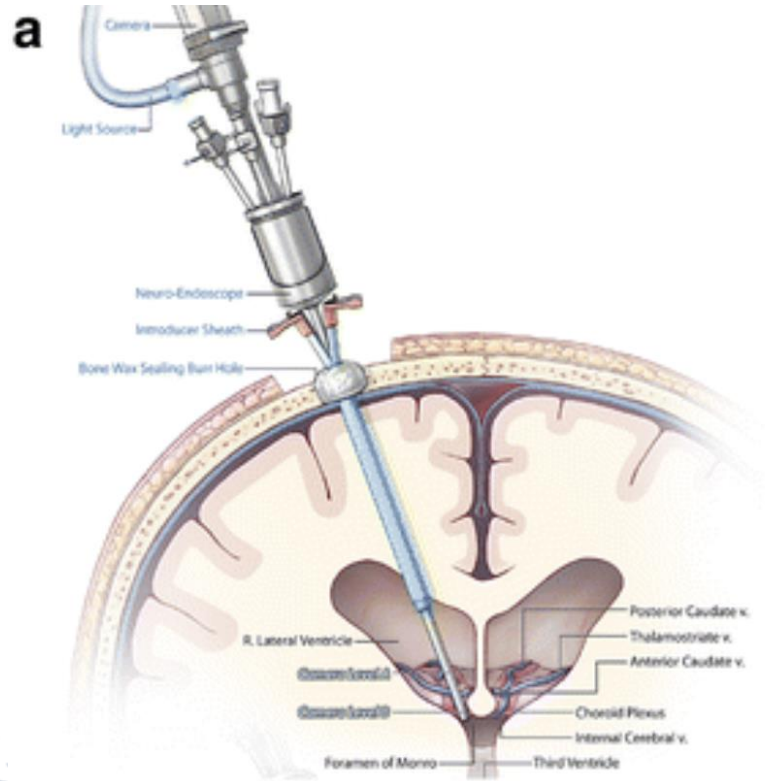
Biopsy

- Allows sampling of lesions to determine identity and behavior
- For multi-bite biopsy, the robot arm allows precise advancement in small steps for sampling different zones of tissue



Ventriculoscapy

- Passing of an endoscope into the ventricles for diagnosis, treatment, etc



Ventriculoscropy

Voice of Garni Barkhoudarian, MD
Director, Adult Hydrocephalus Center

in a 64-year-old man, with progressive headaches
and hydrocephalus with evidence of obstruction

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Play (k)



0:11 / 2:09



HD



Spine Surgery



Spine Surgery

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Thank you!

Additional References

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